
Demand Analysis with interval-valued sales

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Abstract

Most empirical demand studies assume that product sales, and therefore market shares, are fully observed. This assumption is restrictive in digital markets, where downloads or installs are often reported in intervals, as on Steam or the Google Play Store. Interval-valued sales can also arise when only aggregate market sales are observed. This paper studies discrete-choice demand estimation when sales are interval-valued. In a conditional logit model, I show that interval-valued sales imply bounds on mean utilities and lead naturally to moment inequalities and partial identification of demand parameters. I combine two inference approaches: a max-statistic test for moment inequalities and a support-function approach that delivers projected identified intervals for individual coefficients, with smoothing for discrete instruments. Simulations show that replacing each interval by its midpoint can lead to biased and overly precise estimates. I then provide two empirical illustrations. In the Steam application, using data on recent casual action-adventure video games, the partial-identification confidence region allows economically consistent negative values of the price coefficient, while midpoint-based confidence intervals rule them out. In the Google Play Store application, using free puzzle apps released in 2019, midpoint-based estimation suggests several significant effects. The partial-identification results are more cautious: at the estimated identified set level, only the intercept and log rating-count-per-day coefficient rule out zero, and the identified set reveals trade-offs across parameters that a single midpoint estimate cannot show.

Keywords Discrete choice demand; interval data; moment inequalities; support function; digital markets

1. INTRODUCTION

Existing methods in discrete-choice demand analysis (Berry, 1994; Berry and Haile, 2014; Berry et al., 1995, 2004; Nevo, 2001) have mainly assumed that market shares are point-observed, ruling out the possibility of having censored sales. Censored sales occur when sales are reported in intervals. Demand estimates in such a context are related to censored data regression.

A censored data issue occurs when the “true” value of a variable of interest is not known for some observations and is replaced with some partial information. When those unknown true values are all below (resp. above) a certain known threshold and are all replaced by a constant value, they are said to be left (resp. right) censored. When the censored variable is a continuous one, many empirical works have relied on Tobit regression. This method has been particularly popular in wage analysis, as the potential wages for unemployed people are left-censored and replaced by 0. Tobit regression relies on the main assumption that the distribution of the unobservables is known (and is gaussian). However, this assumption is not common in studies on discrete choice demand where analysts have an interest in estimating the distribution of the unobservables, i.e. the unobserved product attributes (Berry et al., 1995).

In addition to left and right-censored data, data can also be censored when the unknown true value is replaced by a closed interval where it almost certainly belongs. Interval data regressions cover such cases. In some income analyses, household incomes are often interval-reported in surveys. In that case, one simple approach often used is to take a “representative” point within the interval, generally the mid-point. However, the underlying assumption is that the midpoint is fairly representative of the whole interval. This is problematic when the censored variable has an asymmetric distribution around the midpoint of the interval. This is typically true for income and sales whose distributions are known to be skewed. A second simple approach - an extension of the Tobit model to the interval data regression - would be to address this interval data issue by assuming a known distribution of the unobservables and then running regressions based on the likelihood function to point identify the parameters of interest (Bettin and Lucchetti, 2012; Stewart, 1983). However, in demand analysis, as mentioned above, it is not convenient to make such an assumption as the distribution of these unobservables is insightful to researchers. Instead, it is common to make the less restrictive assumption of conditional mean independence of demand shocks and some instruments (Berry and Haile, 2014; Berry et al., 1995).

Analysts can carry out regression procedures that do not require to assume that they know the distribution of demand shocks and that rely on partial identification. Partial observation of sales inherently leads to partial identification of demand parameters when no restrictions that can sustain point identification are added.

Interval-valued sales can occur in various cases. In the recent and fast-growing app sector, app providers often disclose their sales data - represented by the volume of downloads or installs - in intervals. These intervals are in the following format for apps that are on the Play Store platform: 100+, 1000+, 10k+, 100k+, 1M+, 10M+, etc. As another example, downloads of digital video games on Steam are accessible through SteamPy and Steam API in the form of interval values. Interval-valued sales can be derived when aggregate sales are

recorded by the empiricist while product-specific characteristics are observed. In this case, the quantity of sales for each product in a given market is not observed, but is known to ultimately be upper-bounded by the market aggregate sales.

Since interval-valued demand may lead to moment inequalities without point identification, this paper is an attempt to apply partial identification methods to discrete choice demand estimation when sales are observed in interval values, including sales for the outside good. I implement and compare two complementary inference strategies. The first is the CCK max-statistic test of Chernozhukov et al., 2019, a general moment-inequality method that is robust to high-dimensional instrumentation. The second exploits the convex-geometric structure of the identified set: building on Bontemps et al., 2012 (BMM), I estimate the support function of the identified set to obtain pointwise confidence intervals for individual coefficients via the BMM Gaussian test statistic. To handle the non-differentiability of the support function that arises when instruments have mass points (a common situation in demand estimation, where dummies and indicator-style instruments are routine), I apply the instrument-smoothing technique of Chandrasekhar et al., 2012 (CCMS). I provide evidence of a potential bias when using the midpoint of the interval as the true value for sales and performing point identification. From an illustration with data on casual video game apps from the action-adventure genre on Steam, I was able to compute a confidence set that allows a negative price coefficient to be considered plausible, whereas this is ruled out by the confidence intervals obtained by considering the midpoint as the actual sales value. From a second illustration with data on free puzzle apps released on Google Playstore, the midpoint-based estimation suggested several significant effects, some of which were economically non-consistent. Meanwhile, the partial-identification results are more cautious: at the estimated identified-set level, only the intercept and log rating-count-per-day coefficient rule out zero, and the wider set reveals trade-offs across parameters that a single midpoint estimate cannot show.

The rest of the paper is organized as follows. Section 2 briefly reviews the literature on moment inequalities and partial identification and their application in empirical industrial organization. Section 3 presents the model of discrete choice demand under the conditional logit setting with partial observation of sales, and describes the inference method carried out. Section 4 then implements this method in simulations and compares it with the point estimate obtained by using the midpoint of the interval as the actual sales value. Finally, the section 5 illustrates this method using data from recent casual video games in the action-adventure genre.

2. LITERATURE

From old topics - such as the measurement error leading to bounds in the estimation of coefficients in linear models (Duncan and Davis, 1953; Fréchet, 1951; Frisch, 1934; Marschak and Andrews, 1944) - to recent issues of identification and inference (D. W. Andrews and Shi, 2013; Beresteanu and Molinari, 2008; Bontemps et al., 2012; Chernozhukov et al., 2007, 2013, 2019; Cox and Shi, 2023; Kaido et al., 2019; Manski and Tamer, 2002; Pakes et al., 2015), there are several existing studies in the literature on partial identification. The most contemporary - starting with Chernozhukov et al., 2007 - develop test inversion procedures to obtain the confidence region of parameters based on moment inequalities (D. W. Andrews and Shi, 2013; Chernozhukov et al., 2013, 2019; Cox and Shi, 2023; Kaido et al., 2019; Pakes et al., 2015). The challenge that these studies typically attempt to overcome is to develop an inference strategy with less computational cost, more power and less conservative confidence regions. For a comprehensive literature review, we refer to Canay et al., 2023; Molinari, 2020; Tamer, 2010.

The computation cost-effectiveness sought by some studies (I. Andrews et al., 2023; Cox and Shi, 2023; Kaido et al., 2019; Pakes et al., 2015) consists in reducing the computational load in the case there are several parameters. This is convenient in demand analysis when there are several relevant product attributes to consider. Other studies (Chernozhukov et al., 2019) have the merit to be robust to an increase in the number of moment inequalities, even above the sample size. Since the number of moments increases with the number of instruments, we often have this many moments setting in demand analysis when we use BLP instruments - i.e. rival products' characteristics (Berry et al., 1995). This happens when there are many product characteristics that the researcher considers as relevant to consumers.

Empirical industrial organization economists have used moment inequality restrictions and partial identification for reasons other than partially observed sales. In English auctions, the intrinsic valuations that players assign to the object of the auction are unknown but needed as researchers are interested in the distribution of those private valuations. According to economic theory-based restrictions, the private valuation of a player is lower-bounded by their revealed bid and upper-bounded by the winner's bid. These restrictions are enough to partially identify the distribution of private valuations (Haile and Tamer, 2003; Tamer, 2010).

Another example where moment inequality restrictions and partial identification are used is related to entry games under complete information and pure strategy (Ciliberto and Tamer, 2009; Molinari, 2020). In such games, each firm decides whether to enter the market or not; and the subgame perfect Nash equilibrium (SPNE) can be unique or multiple depending on the firms' payoff shifters. For regions of observable and unobservable payoff shifters where many outcomes are predicted in equilibrium, the economic theory fails to select a

single outcome. Therefore, there is a partial identification of outcomes in such regions. As a consequence, if the researcher is interested in the probability that a specific outcome is realized given the observable payoff shifters, if that outcome is often predicted as part of a multiple equilibrium, the probability of interest cannot be point-identified without additional restrictions. However, Ciliberto and Tamer, 2009 suggests that the probability of interest is lower-bounded by the model-implied probability that the outcome is a unique equilibrium; and upper-bounded by the model-implied probability that the outcome is an equilibrium¹. This leads inherently to moment inequality restrictions and partial identification.

Beyond these economic theory-based restrictions, the revealed preference restrictions widely lead to partial identification with many moment inequalities when they're not associated with an assumption of a known distribution for the unobserved heterogeneity.

A parallel strand of the partial identification literature exploits the convex-geometric structure of the identified set when moment restrictions are affine in the parameter. Beresteanu and Molinari, 2008 use random-set theory to characterize the identified set via its support function and derive the asymptotic distribution of support function estimators. Bontemps et al., 2012 extend this framework, provide closed-form expressions for the support function (including the case of supernumerary moment restrictions), develop Sargan-type tests for non-emptiness, and propose a test of $H_0 : \theta_0 \in \Theta_I$ based on a perturbation device that handles non-uniqueness at kinks of the identified set Θ_I . Chandrasekhar et al., 2012 address the non-differentiability that arises in the presence of discrete instruments through an instrument-smoothing technique. Kaido and Santos, 2014 and Kaido et al., 2019 further develop the asymptotic theory of support function estimators and propose efficient confidence intervals for projections. More recently, Gafarov, 2025 reformulate the support function estimation problem as a regularized quadratic program, providing uniformly valid inference for polygon-shaped identified sets without requiring instrument smoothing. The support function-based approach is computationally appealing because it transforms a multidimensional inference problem into the estimation of a function on the unit sphere, and yields closed-form asymptotic Gaussian distributions for the test statistics under appropriate regularity conditions.

The present article is not the first study of demand when sales are observed in formats that do not meet the requirements of most existing discrete choice demand models. Gandhi et al., 2023 dealt with many zero market shares and developed an inference strategy that relies on moment inequalities while still using point identification. Indeed, their moment inequalities derived from the deviations between observed sales and equilibrium demand become asymptotic equalities. However, this asymptotic point identification cannot be applied when all observations have interval-valued market shares, as is the case in this study. Therefore, I contribute to censored demand regression models by applying recent developments in the

¹See Molinari, 2020 for more details

literature of partial identification from moment inequalities (Chernozhukov et al., 2019) and from convex-geometric support function estimation (Bontemps et al., 2012; Chandrasekhar et al., 2012) to estimate demand when sales are observed in intervals, including sales for the outside option.

3. MODEL

3.1. Setting

In a market $t \in \{1, \dots, T\}$ with J_t products, the payoff that the consumer i gains from getting product j is

$$U_{ijt} = X'_{jt}\theta + \xi_{jt} + \varepsilon_{ijt} \equiv \delta_{jt} + \varepsilon_{ijt} \quad (1)$$

$X_{jt} \in \mathbb{R}^{\mathcal{K}}$ is the observed product characteristics (including endogenous characteristics such as prices) that affect the mean utilities. $\xi_{jt} \in \mathbb{R}$ represents the unobserved product heterogeneity - known to the consumer. Their distribution is assumed to be unknown to the analyst. δ_{jt} refers to the mean utility from product j in market t . $\varepsilon_{ijt} \in \mathbb{R}$ is the unobserved consumer preference. The analyst observes the data $\{W_t\}_t$, where $W_t = \left(\{S_{jtL}, S_{jtU}, X_{jt}, Z_{jt}\}_{j \in J}, M_{Lt}, M_{Ut}\right)$, where S_{jtL}, S_{jtU} refer to the lower and upper bounds for S_{jt} , the market sales quantity for good j and $Z_{jt} \in \mathbb{R}^{\mathcal{L}}$ is a vector of instruments. M_{Lt}, M_{Ut} refer to the lower and upper bounds for M_t , the market size.

The main assumptions are stated as follows.

Assumption 1. *Data Independence (DI): Demand shocks $\{\xi_{jt}\}_{j,t}$ are independent across j, t .*

Assumption 2. *Conditional Logit (CL): Consumer shocks $\{\varepsilon_{ijt}\}_{i,j,t}$ are i.i.d type-1 extreme value errors.*

Assumption 3. *Partial Observation (PO) of sales: The econometrician observes random variables $M_{tL}, M_{tU}, S_{jtL}, S_{jtU} \forall j$, such that:*

$$Prob \left(0 < S_{jtL} \leq S_{jt} \leq S_{jtU}, \forall j; \sum_{j \in J_t} S_{jtU} < M_{tL} \leq M_t \leq M_{tU} \right) = 1$$

Assumption 4. *Conditional Mean Independence (CMI) of demand shocks: $E(\xi_{jt}|Z_{jt}) = 0$*

Assumption 5. *Full Rank condition: $\text{rank}(E(Z_{jt}Z'_{jt})) = \mathcal{L}$ and $\text{rank}(E(Z_{jt}X'_{jt})) = \mathcal{K} \leq \mathcal{L}$*

While assumptions 2, 3, 4, 5 are useful for the identification, the assumption 1 is useful for the inference. Assumptions 4, 5 ensure point identification in the particular case a single selection $\{S_{jt}\}_j, M_t$ is observed *a.s.*

Assumption 2 implies a conditional logit specification for the demand function $\sigma_j(\delta_t)$, where $\delta_t = (\delta_{1t}, \dots, \delta_{J_t t})$ is the vector of mean utilities across products in market t . The inverse $\sigma_j^{-1}(\mathbf{s}_t)$ is the mean utility δ_{jt} obtained by inverting the vector of market shares $\mathbf{s}_t \equiv \frac{S_t}{M_t}$ (see Berry, 1994; Berry et al., 1995 for the inversion). The inverted demand is $\sigma_j^{-1}(\mathbf{s}_t) = \text{Log}(S_{jt}) - \text{Log}(M_t - \sum_{j=1}^J S_{jt})$. Since δ_{jt} is also a function of product attributes ($\delta_{jt} = X'_{jt}\theta + \xi_{jt}$), these two expressions for the mean utility - namely $\sigma_j^{-1}(\mathbf{s}_t)$ and $X'_{jt}\theta + \xi_{jt}$ - lead to the following demand equation:

$$\text{Log}(S_{jt}) - \text{Log}(M_t - \sum_{j=1}^J S_{jt}) = X'_{jt}\theta + \xi_{jt} \quad (2)$$

Assumption 3 states that the observed random variables $S_{jtb}, M_{tb}, b \in \{L, U\}$, are bounds for sales and market size. It also states that sales are bounded away from zero and the market size is bounded away from the total sales. These additional bounds are required so that the mean utilities obtained from market sales inversion have finite lower and upper bounds.

3.2. Identification and Estimation method

3.2.1 The Identified set under partial observation of sales

In the following, I describe a pseudo-market share inversion. This is a market share inversion - from Berry, 1994; Berry et al., 1995 - extended to a setting of interval-valued sales with a focus on the conditional logit model.

From assumption 3, we observe bounds for sales and bounds for market size. We know that $\delta_{jt} = \text{Log}(S_{jt}) - \text{Log}(M_t - \sum_{l \in J_t} S_{lt})$ is strictly increasing in $\{S_{lt}\}_l$ and strictly decreasing in M_t . Hence, the Lower bound

$$\begin{aligned} \delta_{jtL} &\equiv \inf_{\{M_t, S_{1t}, \dots, S_{J_t t}\}} \text{Log}(S_{jt}) - \text{Log}(M_t - \sum_{l \in J_t} S_{lt}) \\ s|t \quad &M_{tL} \leq M_t \leq M_{tU} \\ &S_{tL} \leq S_{lt} \leq S_{tU} \quad \forall l \in \{1, \dots, J_t\} \\ &\sum_{l \in J_t} S_{lt} \leq M_t \end{aligned}$$

is reached at

$$\delta_{jtL} = \text{Log}(S_{jtL}) - \text{Log}(M_{tU} - \sum_{l \in J_t} S_{ltL}) \quad (3)$$

which is well-defined since $S_{jtL}, M_{tU} - \sum_{l \in J_t} S_{ltL} > 0$ a.s. by assumption 3.

The upper bound

$$\begin{aligned}\delta_{jtU} &\equiv \sup_{\{M_t, S_{1t}, \dots, S_{J_t t}\}} \text{Log}(S_{jt}) - \text{Log}(M_t - \sum_{l \in J_t} S_{lt}) \\ s|t \quad M_{tL} &\leq M_t \leq M_{tU} \\ S_{ltL} &\leq S_{lt} \leq S_{ltU} \quad \forall l \in \{1, \dots, J_t\} \\ \sum_{l \in J_t} S_{lt} &\leq M_t\end{aligned}$$

is reached at

$$\delta_{jtU} = \text{Log}(S_{jtU}) - \text{Log}(M_{tL} - \sum_{l \in J_t} S_{ltU}) \quad (4)$$

which is well-defined since $S_{jtU}, M_{tL} - \sum_{l \in J_t} S_{ltU} > 0$ *a.s.* by assumption 3.

Now that the bounds for mean utilities have been established under the conditional logit setting, we characterize the identified set. Denotes $\Theta \subset \mathbb{R}^{\mathcal{K}}$ the parameter space. Under assumptions 2, 3 and 4, the identified set as defined by conditional moment inequalities is given by:

$$\Theta_I = \{\theta \in \Theta : E[\delta_{jtL}|Z_{jt}] \leq E[X_{jt}|Z_{jt}]'\theta \leq E[\delta_{jtU}|Z_j], \quad \forall Z_{jt}\} \quad (5)$$

Considering that I apply an inference method based on unconditional moment inequalities, as the one proposed by Chernozhukov et al., 2019, the inference is actually about an outer set close to Θ_I and defined by unconditional moment restrictions as follows.

$$\Theta_{\mathcal{G}} = \{\theta \in \Theta : E[\eta_b g(Z_{jt})(X'_{jt}\theta - \delta_{jtb})] \leq 0 \quad \forall g \in \mathcal{G}, \forall b \in \{L, U\}\} \quad (6)$$

Where $\eta_b = -1$ if $b = L$, $\eta_b = 1$ if $b = U$. $\mathcal{G}$ is a convenient countable set of non-negative instrumental functions. Although Θ_I and $\Theta_{\mathcal{G}}$ are identical only under strong restrictions imposed on $\mathcal{G}$, I consider $\Theta_I \approx \Theta_{\mathcal{G}}$. In the present study, $\mathcal{G}$ is built as hypercube indicator functions. This way of obtaining instrumental functions was proposed and shown by D. W. Andrews and Shi, 2013 as suitable to keep information contained in conditional moment restrictions when transforming them into unconditional ones. When there are continuous variables in Z_{jt} , the number of hypercube indicator functions obtained using D. W. Andrews and Shi, 2013 weakly grows with the sample size. Henceforth, an inference method that accounts for this increasing relationship between the size of instrumental functions and the sample size would be appropriate. The definitions 5 and 7 of the identified set are identical when Z_{jt} is discrete and finite.

Let's note that the identified set $\Theta_{\mathcal{G}}$ actually depends on the data generating process (DGP). To make that explicit, denotes $W_{jt} = (X_{jt}, \delta_{jtL}, \delta_{jtU})$ the per-product observed data, drawn from probability distribution P , common across products. P is from a family $\mathcal{P}$ that

will be defined later. $m_v(\theta, W_{jt}) = \eta_b g(Z_{jt})(X'_{jt}\theta - \delta_{jtb})$, $v \in \mathcal{V}$, are moment functions, where $v = (b, g)$ and $\mathcal{V} = \{L, U\} \times \mathcal{G}$. The identified set under the DGP P is expressed as:

$$\Theta_{\mathcal{G}}(P) = \{\theta \in \Theta : E_P[m_v(\theta, W_{jt})] \leq 0 \quad \forall v \in \mathcal{V}\} \quad (7)$$

3.2.2 Confidence region

The goal is to estimate the confidence region $CR(1 - \alpha)$ of size $1 - \alpha$ for $\theta \in \Theta_{\mathcal{G}}(P)$ from data and for distributions P that may have generated the observed data. Methods developed in recent studies in partial identification generally proceed by inverting a test of moment (in)equalities:

$$\begin{cases} H_{0,P}(\theta) : \max_{v \in \mathcal{V}} E_P(m_v(\theta, W_{jt})) \leq 0 \\ H_{1,P}(\theta) : \max_{v \in \mathcal{V}} E_P(m_v(\theta, W_{jt})) > 0 \end{cases} \quad (8)$$

The common inference strategy consists of running the test for each of the parameter values in a grid of parameter space, and then gathering values that fail to reject $H_{0,P}$ as part of the confidence region. In this paper, I implement two complementary inference strategies. The first is the CCK max-statistic test (Chernozhukov et al., 2019), which is robust to many moment inequalities relative to the sample size, but does not exploit the convex-geometric structure of $\Theta_{\mathcal{G}}(P)$ and does not provide direct inference for individual coordinates of the parameter vector θ . The second relies on direct estimation of the support function of $\Theta_{\mathcal{G}}(P)$ following Bontemps et al., 2012 (BMM hereafter) and Beresteanu and Molinari, 2008, and uses the smoothing technique of Chandrasekhar et al., 2012 (CCMS hereafter) to handle the discrete instruments that arise in our application. It also gives direct inference of individual elements of θ . The two approaches use different characterizations of the identified sets and address different aspects of inference: CCK considers moment inequality-based characterization of $\Theta_{\mathcal{G}}(P)$ and provides a robust joint confidence region; while BMM consider support function-based and sharp geometric description of $\Theta_{\mathcal{G}}(P)$ and can be used to obtain direct confidence intervals for individual coordinates of θ .

The test statistics implemented in the moment-inequality literature are typically from the following categories:

- the max statistic: maximum across standardized average moments (I. Andrews et al., 2023; Canay et al., 2023; Chernozhukov et al., 2019).
- quasi-likelihood ratio statistic (D. W. Andrews and Barwick, 2012; Canay, 2010; Rosen, 2008).
- modified method of moment statistic (D. W. Andrews and Soares, 2010; Bugni, 2010; Ciliberto and Tamer, 2009).

The support function-based methods of BMM and CCMS are not test statistics in the same family as those above. Rather, they replace the test problem (8) with a direction-by-direction characterization of $\Theta_G(P)$, leading to test statistics with explicit Gaussian limits.

Max-statistic test (CCK) I opt for the max statistic from Chernozhukov et al., 2019 (hence CCK test statistic)²:

$$T_n(\theta) = \max_{v \in \mathcal{V}} \frac{\sqrt{n} \bar{m}_v(\theta)}{\hat{\sigma}_{n,v}(\theta)}$$

Where:

- $\bar{m}_v(\theta) = \frac{1}{n} \sum_{j,t} m_v(\theta, W_{jt})$ is the average of the l th moment.
- $\hat{\sigma}_{n,v} = \sqrt{\frac{1}{n} \sum_{j,t} (m_v(\theta, W_{jt}) - \bar{m}_v(\theta))^2}$ is its standard deviation.
- n represents the sample size ($n = \sum_{t=1}^T J_t$).

This max-type statistic has some robustness properties as its critical value increases slowly with the number of moment inequalities. As a result, it remains suitable in models with large moment inequalities - relative to sample size n . Moreover, it has some power as it increases weakly with the value of each moment. See Canay et al., 2023; Chernozhukov et al., 2019 for more on the test statistics and on the suitable properties of the max statistic.

Chernozhukov et al., 2019 propose different test methods to compute critical values: a self-normalized (SN) critical value, which is based on a pivotal statistic (a standard normal); and two bootstrap-based critical values: the multiplier bootstrap (MB) and the empirical bootstrap (EB). They propose three variants for each of these three methods: a one-step variant without moment selection; a two-step variant that discards uninformative moment inequalities that are statistically not binding; and a three-step variant where the authors propose a new selection of “weakly informative inequalities that are potentially binding but do not provide first-order information” (Chernozhukov et al., 2019). To conciliate the computation effectiveness of the SN method with the power and low conservativeness of the bootstrap methods, they also propose hybrid methods.

In this paper, in the simulation exercises carried out in section 4, I am using the two-step hybrid method with multiplier bootstrap critical values in the second step. This choice is motivated by its conciliatory advantage between the computational effectiveness of the SN method in the first step and the power of the bootstrap method in the second step. In the illustration presented in section 5, which uses data on a small number of products from the Steam platform, I consider only a few product attributes, which solves the computational burden and allows me to use the two-step method with multiplier bootstrap critical values in both the first and second steps.

The algorithm from Chernozhukov et al., 2019 in the two-step MB method is described

²In a context of data independence, i.e. when the error term ξ_{jt} is independent across observations j, t

as follows:

Step 0: Generate B vectors $\{\{U_{jt}^1\}_{j,t}, \dots, \{U_{jt}^b\}_{j,t}, \dots, \{U_{jt}^B\}_{j,t}\}$ of independent standard normal random variables independent of the data $\{W_{jt}\}_{j,t}$.

Step 1: Moment selection using MB critical value

1. Construct the multiplier bootstrap test statistic for the first step critical value:

$$T_n^b(\theta) = \max_{v \in \mathcal{V}} \frac{\sqrt{n} \tilde{m}_v^b(\theta)}{\hat{\sigma}_{n,v}(\theta)} \quad \forall b \in \{1, \dots, B\}$$

where

$$\tilde{m}_v^b(\theta) = \frac{1}{n} \sum_{j,t} U_{jt}^b (m_v(\theta, W_{jt}) - \bar{m}_v(\theta))$$

2. Compute the related first step critical value $c_n^1(\theta, \beta_n)$ of size β_n , as the $1 - \beta_n$ -quantile of $\{T_n^b(\theta)\}_{b=1}^{b=B}$; where β_n is a chosen such that $0 < \beta_n < \alpha/2$ ³
3. Get an estimated set of informative moments

$$\hat{\mathcal{V}} = \{v \in \mathcal{V} : \frac{\sqrt{n} \bar{m}_v(\theta)}{\hat{\sigma}_{n,v}(\theta)} > -2c_n^1(\theta, \alpha)\}$$

Step 2: Compute the critical value for the test 8.

1. Construct the multiplier bootstrap test statistic for the second step critical value:

$$Y_n^b(\theta) = \begin{cases} \max_{v \in \hat{\mathcal{V}}} \frac{\sqrt{n} \tilde{m}_v^b(\theta)}{\hat{\sigma}_{n,v}(\theta)} & \text{if } |\hat{\mathcal{V}}| > 0 \\ 0 & \text{if } |\hat{\mathcal{V}}| = 0 \end{cases}$$

2. Compute the critical value $c_n(\theta, \alpha)$ of size α for the test 8, as the $(1 - \alpha + 2\beta_n)$ -quantile of $\{Y_n^b(\theta)\}_{b=1}^{b=B}$.
3. Conclude: do not reject $H_{0,P}(\theta)$ if $T_n(\theta) \leq c_n(\theta, \alpha)$.

The algorithm for the two-step hybrid method with MB in the second step is similar to the previous algorithm except that there is no step 0 and the first step critical value c_n^1 is from the SN method and has a closed-form expression:

$$c_n^1(\theta, \beta_n) \equiv c_n^1((\beta_n)) = \frac{\Phi^{-1}(1 - \alpha/|\mathcal{V}|)}{\sqrt{1 - \Phi^{-1}(1 - \alpha/|\mathcal{V}|)^2/n}}$$

³Chernozhukov et al., 2019 suggest $\beta_n = \alpha/50$ based on their simulation exercices. Canay et al., 2023, pp 13, also mention this rule of thumb.

where Φ is the cumulative distribution function of a standard normal random variable, and its inverse Φ^{-1} is the corresponding quantile function.

Chernozhukov et al., 2019 also propose a version of the multiplier bootstrap, called block multiplier bootstrap, suitable for data dependent cases. In this test procedure, The test statistic is the non-studentized version of $T_n(\theta)$ and the bootstrap statistic used to compute the critical value is a version of T_n^b where the random variables U_{jt}^b are common within blocks of observations.

The level $(1 - \alpha)$ -confidence region for $\theta \in \Theta_{\mathcal{G}}(P)$ is defined by the set $CR_n(1 - \alpha)$ of all θ from the parameter space for which $H_{0,P}(\theta)$ is not rejected by the test 8. More precisely,

$$CR_n(1 - \alpha) = \{\theta \in \Theta : T_n(\theta) \leq c_n(\theta, \alpha)\}$$

where $T_n(\theta)$ is the test statistic and $c_n(\theta, \alpha)$ is the corresponding critical value such that

$$\theta \in \Theta_{\mathcal{G}}(P) \implies \text{plim} \text{Prob}(\theta \in CR_n(1 - \alpha)) \geq 1 - \alpha$$

Hence, we can assert with a probability weakly greater than $1 - \alpha$ asymptotically that a given parameter from the identified set is an element of $CR(1 - \alpha)$. Chernozhukov et al., 2019 provide regularity conditions - related to appropriate variations and bounds in population moments of studentized transformations of $m_v(\theta, W_{jt})$ - under which $CR_n(1 - \alpha)$ is asymptotically honest with a polynomial rate. $CR_n(1 - \alpha)$ is asymptotically honest - with a polynomial rate - to a suitable sequence $\mathcal{P}_n$ of classes of distributions P of random vectors W_{jt} if $CR_n(1 - \alpha)$ has a coverage being correct uniformly in $\mathcal{P}_n$. Denotes

$$\begin{aligned} \mu_{v,P}(\theta) &= E_P[m_v(\theta, W_{jt})], & \sigma_{v,P}^2(\theta) &= E_P[(m_v(\theta, W_{jt}) - \mu_{v,P}(\theta))^2], \\ Z_{v,P}(\theta, W_{jt}) &= \frac{m_v(\theta, W_{jt}) - \mu_{v,P}(\theta)}{\sigma_{v,P}(\theta)}, & M_{k,P}(\theta) &= \max_{v \in \mathcal{V}} (E_P[|Z_{v,P}(\theta, W_{jt})|^k])^{1/k}, \quad k \in \{3, 4\}, \\ B_P(\theta) &= \left(E_P \left[\max_{v \in \mathcal{V}} Z_{v,P}^4(\theta, W_{jt}) \right] \right)^{1/4}. \end{aligned}$$

Conditions for validity of the confidence set $CR_n(1 - \alpha)$ rely on bounds of these expressions.

Assumption 6. *Regularity conditions from Chernozhukov et al., 2019 (CCK)*

For a given distribution P and $\forall \theta \in \Theta_{\mathcal{G}}(P)$;

- (i) $E_P[m_v^2(\theta, W_{jt})] < +\infty$ and $\sigma_{v,P}^2(\theta) > 0$;
- (ii) $\exists 0 < c_1 < 1/2$ and $C_1 > 0$, two constants such that:

$$\max\{M_{3,P}^3(\theta), M_{4,P}^2(\theta), B_P(\theta)\}^2(\theta) \log^{7/2}(|\mathcal{V}| \times n) \leq C_1 n^{1/2-c_1};$$

Condition (i) in Assumption 6 ensures that the moment functions $m_v(\theta, W_{jt})$ have finite values and vary in the population. This is the usual regularity condition given when there is point-identification. It is needed so that objects in (ii) are well-defined. Condition (ii) imposes some bounds in the population moments of moment functions.

Remark – These population moments are bounded above by an expression that diverges slowly. Note that $B_P(\theta) \geq M_{4,P}(\theta) \geq M_{3,P}(\theta) \geq 1$. Moreover, if standardized random variables $Z_{v,P}(\theta, W_{jt})$ are all bounded by a constant $C_P(\theta)$ almost surely, then $C_P(\theta) \geq B_P(\theta)$ (see Chernozhukov et al., 2019, in their introduction of section 4.).

The following proposition comes directly from *Theorems 4.3, 4.4, 4.5 and A.1* in Chernozhukov et al., 2019.

Proposition 1. *Let $\mathcal{P}$ be the family of distributions P that satisfy assumption ?? with common constants C_1, c_1 . Moreover, suppose that $\log(1/\beta_n) \leq C_1 \log(n)$. Then there exists positive constants c, C depending only on α, c_1, C_1 such that:*

$$\inf_{P \in \mathcal{P}} \inf_{\theta \in \Theta_0(P)} P(\theta \in CR_n(1 - \alpha)) \geq 1 - \alpha - Cn^{-c}$$

Proof —

Assumptions in Proposition 1 satisfy those of *Theorems 4.4, 4.5* in Chernozhukov et al., 2019 for each $P \in \mathcal{P}$, $\theta \in \Theta_{\mathcal{G}}(P)$. Under their *Theorems 4.4, 4.5*, there exist constants $0 < c(P, \theta) < 1/2$ and $C(P, \theta) > 0$ such that $\theta \in \Theta_{\mathcal{G}}(P) \implies P(T_n(\theta) > c_n(\theta, \alpha)) \leq \alpha + C(P, \theta)n^{-c(P, \theta)}$; i.e. $P(T_n(\theta) \leq c_n(\theta, \alpha)) > 1 - \alpha - C(P, \theta)n^{-c(P, \theta)}$. Taking (c, C) such that $Cn^{-c} = \min_{\theta \in \Theta_{\mathcal{G}}(P), P \in \mathcal{P}} C(P, \theta)n^{-c(P, \theta)}$ and then taking the infimum of $P(T_n(\theta) \leq c_n(\theta, \alpha))$ across $\theta \in \Theta_{\mathcal{G}}(P)$ and across $P \in \mathcal{P}$, we get the result. $\square$

Proposition 1 states that under some regularity conditions, $CR_n(1 - \alpha)$ is asymptotically honest - with a polynomial rate - to a suitable family $\mathcal{P}$ of distributions of W_{jt} . Indeed, asymptotically, the minimum coverage rate guaranteed is weakly greater than $1 - \alpha$. The error margin of that coverage rate (Cn^{-c}) is polynomial in n and vanishes as n goes to $+\infty$.

Support function-based test (BMM) While the CCK test inversion above is general and robust to many moment inequalities, it does not exploit the convex-geometric structure of the identified set. Bontemps et al., 2012 (BMM) provide an alternative inference strategy that explicitly leverages the convexity of $\Theta_{\mathcal{G}}(P)$ via its support function. Chandrasekhar et al., 2012 (CCMS) develop similar approaches to identify linear approximations of interval-valued functions when their bounds are estimated non-parametrically. They also address the

exposed-face and kink issues related to discrete instruments by adding a smooth random variable to those instruments.

The identified set as a convex set. Under the assumptions of section 3, the identified set $\Theta_{\mathcal{G}}(P)$ is convex and bounded.⁴ A convex bounded set is characterized by its *support function*, defined for every direction q on the unit sphere $S^{\mathcal{K}-1} = \{q \in \mathbb{R}^{\mathcal{K}} : \|q\| = 1\}$ as

$$\delta^*(q \mid \Theta_{\mathcal{G}}(P)) = \sup_{\theta \in \Theta_{\mathcal{G}}(P)} q' \theta.$$

Geometrically, $\delta^*(q \mid \Theta_{\mathcal{G}}(P))$ is the signed distance from the origin to the supporting hyperplane of $\Theta_{\mathcal{G}}(P)$ with outward normal q . Membership in $\Theta_{\mathcal{G}}(P)$ admits an equivalent direction-by-direction characterization,

$$\theta_0 \in \Theta_{\mathcal{G}}(P) \iff q' \theta_0 \leq \delta^*(q \mid \Theta_{\mathcal{G}}(P)) \text{ for all } q \in S^{\mathcal{K}-1}.$$

Inference on θ_0 then reduces to inference on a scalar pivotal quantity in the worst-case direction.

Closed-form expression of the support function. Let $\Sigma = (E[Z_{jt}X'_{jt}])^{-1}$, $z_q = q' \Sigma Z_{jt}$, and $w_q = \delta_{jtL} + \mathbf{1}\{z_q > 0\}(\delta_{jtU} - \delta_{jtL})$. When the dimension l of instruments equals the dimension k of regressors (just-identified case), Proposition 2 of Bontemps et al., 2012 gives

$$\delta^*(q \mid \Theta_{\mathcal{G}}(P)) = E[z_q w_q].$$

When $\mathcal{L} > \mathcal{K}$ (supernumerary instruments), the support function is obtained by minimizing the support function of an augmented unconstrained set $\Theta_{\mathcal{G}}^U(P) \subset \mathbb{R}^{\mathcal{L}}$ over the supernumerary directions (Proposition 6 of Bontemps et al., 2012):

$$\delta^*(q \mid \Theta_{\mathcal{G}}(P)) = \inf_{\lambda \in \mathbb{R}^{\mathcal{L}-\mathcal{K}}} \delta^*((q, \lambda) \mid \Theta_{\mathcal{G}}^U(P)).$$

The minimizer $\lambda_m(q)$ is the direction in the supernumerary subspace that makes the projection of $\Theta_{\mathcal{G}}^U(P)$ onto the hyperplane $\gamma = 0$ tangent to $\Theta_{\mathcal{G}}(P)$. The empirical estimator $\hat{\delta}_n^*(q \mid \Theta_{\mathcal{G}}(P))$ replaces population expectations with sample averages and solves the inner minimization on the empirical augmented support function. Existence of the inner argmin requires a non-emptiness condition that generalizes the usual Sargan overidentification condition (BMM Proposition 5).

⁴Convexity follows from the linearity of moment conditions in θ and the convexity of the admissible set $[\delta_{jtL}, \delta_{jtU}]$ for δ_{jt} . Boundedness follows from Assumption 5.

Asymptotic Gaussianity of the empirical support function. The cornerstone of BMM inference is that the empirical support function is, direction by direction, asymptotically Gaussian.

Proposition 2. *Suppose the regularity conditions of Bontemps et al., 2012 (Propositions 9 and 12) hold: finite cross-moments with at least $2 + \gamma$ uniform bounds for some $\gamma > 0$, finitely many mass points in the distribution of Z_{jt} , Assumption D (differentiability of $\delta^*(\cdot \mid \Theta_{\mathcal{G}}(P))$), and, in the supernumerary case, Condition S (the inner minimizer $\lambda_m(q)$ lies in a compact set). Then, for every fixed direction $q \in S^{\mathcal{K}-1}$,*

$$\sqrt{n} \left(\hat{\delta}_n^*(q \mid \Theta_{\mathcal{G}}(P)) - \delta^*(q \mid \Theta_{\mathcal{G}}(P)) \right) \xrightarrow[n \rightarrow \infty]{d} \mathcal{N}(0, V_q),$$

where $V_q = \text{Var}(z_{q,jt} \varepsilon_{q,jt})$, $\varepsilon_{q,jt} = w_{q,jt} - X'_{jt} \theta_q$, and $\theta_q = \Sigma E[Z'_{jt} w_{q,jt}]$ is the population frontier point of $\Theta_{\mathcal{G}}(P)$ in direction q . A consistent estimator of V_q is $\hat{V}_q = \frac{1}{n} \sum_{j,t} (z_{n,q,jt} \varepsilon_{n,q,jt})^2$, and the studentized empirical support function converges to a standard normal:

$$\frac{\sqrt{n} (\hat{\delta}_n^*(q \mid \Theta_{\mathcal{G}}(P)) - \delta^*(q \mid \Theta_{\mathcal{G}}(P)))}{\sqrt{\hat{V}_q}} \xrightarrow[n \rightarrow \infty]{d} \mathcal{N}(0, 1).$$

Proof —

Proposition 9 of Bontemps et al., 2012 establishes, in the just-identified case ($\mathcal{L} = \mathcal{K}$), that the stochastic process $\tau_n(q) = \sqrt{n} (\hat{\delta}_n^*(q \mid \Theta_{\mathcal{G}}(P)) - \delta^*(q \mid \Theta_{\mathcal{G}}(P)))$ converges uniformly in $q \in S^{\mathcal{K}-1}$ to a Gaussian process with covariance kernel $E(z_{q,jt} \varepsilon_{q,jt} \varepsilon_{r,jt} z_{r,jt})$, and that this implies marginally $\tau_n(q) \rightarrow_d \mathcal{N}(0, V_q)$ for each fixed q . Proposition 12 of Bontemps et al., 2012 extends the result to the supernumerary case under Condition S. Consistency of $\hat{V}_q$ for V_q follows from the standard analog principle and Slutsky's lemma; division by $\sqrt{\hat{V}_q}$ gives the standard normal limit. $\square$

Proposition 2 is the engine behind every subsequent inferential statement in this section. It is direction-agnostic: the same Gaussian limit applies whether q is chosen as a canonical direction $\pm e_l$ (in which case it delivers a pivotal statistic for the lower or upper bound of the projected identified interval of coordinate l), or chosen data-adaptively as q_n , the perturbed minimizer described below (in which case it delivers a pivotal test statistic for membership of θ_0 in $\Theta_{\mathcal{G}}(P)$ or in $\partial\Theta_{\mathcal{G}}(P)$). The next two subparagraphs make these two specializations explicit.

Test statistic and critical regions. For each candidate parameter θ_0 , define

$$\tau_{\infty}(q; \theta_0) = \delta^*(q \mid \Theta_{\mathcal{G}}(P)) - q' \theta_0, \quad \tau_n(q; \theta_0) = \hat{\delta}_n^*(q \mid \Theta_{\mathcal{G}}(P)) - q' \theta_0.$$

The minimum of $\tau_\infty(\cdot; \theta_0)$ over the sphere is non-negative iff $\theta_0 \in \Theta_{\mathcal{G}}(P)$, with equality attained when θ_0 lies on the frontier $\partial\Theta_{\mathcal{G}}(P)$. To handle non-uniqueness of the argmin at kinks of $\Theta_{\mathcal{G}}(P)$, BMM Proposition 10 introduces a perturbation,

$$\Psi_{n, a_n^{\text{outer}}}(q; \theta_0) = \tau_n(q; \theta_0) - a_n^{\text{outer}} q' v_{0,n}, \quad v_{0,n} = \frac{\theta_0 - \hat{\theta}_n^*}{\|\theta_0 - \hat{\theta}_n^*\|},$$

where $\hat{\theta}_n^*$ is a consistent estimator of an interior point of $\Theta_{\mathcal{G}}(P)$, and $a_n^{\text{outer}} \rightarrow 0$ with $\sqrt{n} a_n^{\text{outer}} \rightarrow \infty$. The minimizer q_n of $\Psi_{n, a_n^{\text{outer}}}(\cdot; \theta_0)$ converges to a unique direction q_0^* that picks out a specific tangent direction at θ_0 even when $\partial\Theta_{\mathcal{G}}(P)$ has kinks. The BMM test statistic is the studentized empirical τ_n evaluated at q_n :

$$T_n(\theta_0) = \frac{\sqrt{n} \tau_n(q_n; \theta_0)}{\sqrt{\hat{V}_{q_n}}}, \quad \hat{V}_{q_n} = \frac{1}{n} \sum_{j,t} (z_{n,q_n,jt} \varepsilon_{n,q_n,jt})^2,$$

where $\varepsilon_{n,q_n,jt} = w_{n,q_n,jt} - X'_{jt} \hat{\theta}_{q_n,n}$ is the empirical IV residual at direction q_n and $\hat{\theta}_{q_n,n}$ is the empirical frontier point in that direction. Proposition 2 applied at the data-driven direction q_n , combined with the perturbation argument of BMM Proposition 10, yields the following three-regime limit.

Proposition 3. *Under the assumptions of Proposition 2 and the perturbation conditions $a_n^{\text{outer}} \rightarrow 0$, $\sqrt{n} a_n^{\text{outer}} \rightarrow \infty$:*

$$T_n(\theta_0) \xrightarrow[n \rightarrow \infty]{d} \mathcal{N}(0, 1) \quad \text{if } \theta_0 \in \partial\Theta_{\mathcal{G}}(P);$$

$$T_n(\theta_0) \xrightarrow[n \rightarrow \infty]{a.s.} +\infty \quad \text{if } \theta_0 \in \text{int}(\Theta_{\mathcal{G}}(P)); \quad T_n(\theta_0) \xrightarrow[n \rightarrow \infty]{a.s.} -\infty \quad \text{if } \theta_0 \notin \Theta_{\mathcal{G}}(P).$$

Proof —

This is the conclusion of Proposition 10 of Bontemps et al., 2012 once Assumption D and the perturbation conditions are verified in our set-up. The Gaussian regime on $\partial\Theta_{\mathcal{G}}(P)$ follows from Proposition 2 applied at $q_n \rightarrow_p q_0^*$, together with stochastic equicontinuity of τ_n on the unit sphere (Andrews, 1994; cf. BMM Appendix C.2). The divergence to $+\infty$ in the interior and $-\infty$ outside follow because $\tau_\infty(q_0^*; \theta_0)$ is then strictly positive or strictly negative, respectively, and $\sqrt{n}$ -scaling dominates. $\square$

Proposition 3 delivers two complementary tests at asymptotic level α :

- *Test 1* (interior, one-sided): $H_0 : \theta_0 \in \Theta_{\mathcal{G}}(P)$, with rejection region $\{T_n(\theta_0) < N_\alpha\}$ where $N_\alpha = \Phi^{-1}(\alpha)$ (e.g. -1.645 for $\alpha = 0.05$).
- *Test 2* (boundary, two-sided): $H_0 : \theta_0 \in \partial\Theta_{\mathcal{G}}(P)$, with rejection region $\{|T_n(\theta_0)| > N_{1-\alpha/2}\}$ where $N_{1-\alpha/2} = \Phi^{-1}(1 - \alpha/2)$ (e.g. 1.96).

By inverting Test 1, the pointwise BMM confidence region (Proposition 11 of Bontemps

et al., 2012) is

$$\text{CI}_n^\alpha = \{\theta_0 : T_n(\theta_0) \geq N_\alpha\},$$

with asymptotic coverage

$$\liminf_{n \rightarrow \infty} \inf_{\theta_0 \in \Theta_{\mathcal{G}}(P)} P(\theta_0 \in \text{CI}_n^\alpha) = 1 - \alpha,$$

the bound attained at boundary points. For uniform coverage including the point-identified limit where the diameter of $\Theta_{\mathcal{G}}(P)$ shrinks to zero, Proposition 19 of Bontemps et al., 2012 (following Imbens and Manski, 2004 and Stoye, 2009) replaces N_α with a critical value $\tilde{N}_\alpha$ satisfying

$$\Phi\left(\tilde{N}_\alpha + \frac{\sqrt{n} \hat{\Delta}_n}{\sqrt{\hat{V}_{q_n}}}\right) - \Phi(-\tilde{N}_\alpha) = \alpha,$$

where $\hat{\Delta}_n = \max_q [\hat{\delta}_n^*(q \mid \Theta_{\mathcal{G}}(P)) + \hat{\delta}_n^*(-q \mid \Theta_{\mathcal{G}}(P))]$ is the estimated diameter. The critical value $\tilde{N}_\alpha$ smoothly interpolates between the two-sided $\Phi^{-1}((1 + \alpha)/2)$ in the point-identified limit and the pointwise one-sided N_α when $\Theta_{\mathcal{G}}(P)$ is clearly proper.

Smoothing for discrete instruments (CCMS). The asymptotic Gaussianity of $T_n(\theta_0)$ in Proposition 3 – and indeed Proposition 2 itself – requires the support function $\delta^*(\cdot \mid \Theta_{\mathcal{G}}(P))$ to be differentiable (Assumption D). By Lemma 3 of Bontemps et al., 2012, this fails exactly at directions q such that $\Pr(z_q = 0) > 0$, which occurs when instruments have mass points. In demand estimation this is a common situation: some discrete BLP-style instruments and discrete exogenous shifters all carry mass points by construction. In the simulations of section 4, our two instruments Z_{1jt}, Z_{2jt} are drawn from a Bernoulli distribution, so the BMM regularity condition is violated.

Chandrasekhar et al., 2012 address this by perturbing the instruments with a small amount of Gaussian noise:

$$Z_{jt}^{\text{smooth}} = Z_{jt} + \gamma \cdot \eta_{jt}, \quad \eta_{jt} \sim \mathcal{N}(0, I_l),$$

where the smoothing scale

$$\gamma = \frac{\gamma_0}{E[|\eta_{jt}|] (E[|\delta_{jtL}|] + E[|\delta_{jtU}|] + E[|\delta_{jtU} - \delta_{jtL}|])}$$

is calibrated to ensure that the distance between the original and the regularized identified sets, $\Theta_{\mathcal{G}}(P)$ and $\Theta_{\mathcal{G}}^{\text{smooth}}(P)$, is no greater than γ_0 . The smoothed instruments have continuous densities, so the resulting identified set $\Theta_{\mathcal{G}}^{\text{smooth}}(P)$ is strictly convex and its support function fully differentiable, restoring the validity of Propositions 2 and 3 on the smoothed problem.

Projected identified intervals for individual coordinates. A direct specialization of Proposition 2 to the canonical directions $q = \pm e_l$ delivers closed-form bounds and pointwise Gaussian confidence intervals for individual components of θ at no additional cost.

Lemma 1. *Denote e_l the l -th coordinate basis vector. The projection of $\Theta_{\mathcal{G}}(P)$ onto the l -th coordinate is the closed interval*

$$[\theta_l^L, \theta_l^U] = [-\delta^*(-e_l \mid \Theta_{\mathcal{G}}(P)), \delta^*(e_l \mid \Theta_{\mathcal{G}}(P))],$$

with empirical analogs $\hat{\theta}_{l,n}^L = -\hat{\delta}_n^*(-e_l \mid \Theta_{\mathcal{G}}(P))$ and $\hat{\theta}_{l,n}^U = \hat{\delta}_n^*(e_l \mid \Theta_{\mathcal{G}}(P))$. Under the assumptions of Proposition 2,

$$\sqrt{n}(\hat{\theta}_{l,n}^L - \theta_l^L) \xrightarrow[n \rightarrow \infty]{d} \mathcal{N}(0, V_{-e_l}), \quad \sqrt{n}(\hat{\theta}_{l,n}^U - \theta_l^U) \xrightarrow[n \rightarrow \infty]{d} \mathcal{N}(0, V_{e_l}),$$

and the pointwise $1 - \alpha$ confidence intervals for the two bounds are

$$[\hat{\theta}_{l,n}^L \pm z_{1-\alpha/2} \sqrt{\hat{V}_{-e_l}/n}] \quad \text{and} \quad [\hat{\theta}_{l,n}^U \pm z_{1-\alpha/2} \sqrt{\hat{V}_{e_l}/n}].$$

Proof —

The closed-form expression of $[\theta_l^L, \theta_l^U]$ follows from the definition of δ^* : $\theta_l^U = \sup_{\theta \in \Theta_{\mathcal{G}}(P)} e_l' \theta = \delta^*(e_l \mid \Theta_{\mathcal{G}}(P))$ and, symmetrically, $\theta_l^L = -\delta^*(-e_l \mid \Theta_{\mathcal{G}}(P))$. Asymptotic normality of $\hat{\theta}_{l,n}^U$ and $\hat{\theta}_{l,n}^L$ are the marginal Gaussian limits of Proposition 2 at the fixed directions e_l and $-e_l$. The confidence interval for each bound is a direct inversion of the standard normal limit. $\square$

This procedure is computationally efficient relative to inverting a test over a multidimensional grid: it requires only $2\mathcal{K}$ evaluations of the empirical support function rather than a grid search.

Computational implementation. The empirical implementation involves two nested optimizations: an outer minimization of the perturbed objective $\Psi_{n, a_n^{\text{outer}}}$ over $q \in S^{\mathcal{K}-1}$, and an inner minimization over λ in the supernumerary case. Two strategies are available for the outer problem.

A first strategy is to solve the outer minimization directly on the sphere via a constrained nonlinear program – in practice, the SLSQP routine of `nloptr` with multiple starting points (the direction toward $\theta_0 - \hat{\theta}_n^*$, its antipodal, and the $\pm e_l$ axis directions) to avoid local minima. This direct approach delivers the exact minimizer q_n at each candidate θ_0 .

A second, computationally cheaper strategy is to discretize $S^{\mathcal{K}-1}$ – drawing a large but fixed grid of directions on the sphere once, and replacing the outer minimization at each θ_0 by a search over this fixed grid. The computational gain over the direct approach is substantial when the test is inverted over a parameter grid containing many θ_0 , since the

empirical support function (and its variance estimator) can be pre-computed at each grid direction and reused. We adopt different choices in different parts of the paper, motivated by the relative computational burden:

- *Simulations* (Section 4). Direct optimization at every (θ_0, n) pair across 500 Monte Carlo replications and several DGPs is prohibitively expensive. We therefore discretize $S^{\mathcal{K}-1}$ into 2,000 directions sampled uniformly on the sphere and minimize $\Psi_{n, a_n^{\text{outer}}}$ over this fixed grid. In dimension-two pilot experiments, the resulting confidence sets had essentially the same shape and coverage as those obtained by direct optimization.
- *Data applications* (Section 5). The test is evaluated on a single dataset, so we use the direct (continuous) optimization with multiple starting points to compute q_n at each θ_0 .

For the inner minimization over λ in the supernumerary case, we use a spectral projected gradient (SPG) method with a separate regularization parameter a_n^{inner} ensuring that the supernumerary minimizer is uniquely defined. The two perturbation parameters a_n^{inner} and a_n^{outer} play distinct roles: the inner one regularizes the supernumerary minimization, and the outer one resolves potential kink ambiguity in $\partial\Theta_{\mathcal{G}}(P)$. In our simulations we set both to $c \cdot n^{-1/3}$ with $c \in [0.1, 0.5]$, following the rate condition in Bontemps et al., 2012 and the simulation choices of Gafarov, 2025.

The full empirical identified set $\hat{\Theta}_{\mathcal{G}, n}$ can also be traced by sampling random directions $q \in S^{\mathcal{K}-1}$ and computing the corresponding boundary point $\hat{\theta}_{q, n} = \hat{\Sigma}_n n^{-1} \sum_{j,t} Z'_{jt} w_{n,q,jt}$. This provides a visual representation of $\hat{\Theta}_{\mathcal{G}, n}$ that complements the test-inversion approach and is useful for diagnostics.

Connection to the linear-in-parameters demand equation. The conditional logit demand equation (2) can be written compactly as $\delta_{jt} = X'_{jt}\theta + \xi_{jt}$ with $E[\xi_{jt} | Z_{jt}] = 0$ (Assumption 4). Under partial observation, δ_{jt} is observed only as an interval $[\delta_{jtL}, \delta_{jtU}]$. This setup maps directly onto BMM’s incomplete linear moment framework with $y = \delta_{jtL}$, $\bar{y} = \delta_{jtU}$, $x = X_{jt}$, $z = Z_{jt}$, and the implied moment condition

$$E[Z_{jt}(X'_{jt}\theta - \delta_{jt})] = 0 \quad \text{for some } \delta_{jt} \in [\delta_{jtL}, \delta_{jtU}].$$

The BMM characterization does not, on its face, require instrumental functions $g(Z_{jt})$ to enter the moment restriction: the original instruments Z_{jt} themselves are enough to define the identified set and its support function. In our implementation, however, we nonetheless apply the same library $\mathcal{G}$ of instrumental functions used in the CCK construction, working with $g(Z_{jt})$ in place of Z_{jt} in the empirical support function. This enrichment is a strictly free choice in the BMM framework – since each $g(Z_{jt})$ satisfies $E[\xi_{jt}g(Z_{jt})] = 0$ under Assumption 4, it remains a valid instrument and yields an additional, possibly tighter, characterization of $\Theta_{\mathcal{G}}(P)$. Using the same $\mathcal{G}$ also keeps the BMM and CCK identified sets directly comparable,

since they are now defined relative to the same instrument library.

A complementary route was recently proposed by Gafarov, 2025, who provides a regularized quadratic-programming formulation of the support function for polygon-shaped identified sets. This approach is appealing when instruments are discrete (so $\Theta_G(P)$ is a polygon with exposed faces), since it avoids the non-differentiability issue without resorting to instrument smoothing, and delivers uniformly valid inference via closed-form Gaussian critical values. We use BMM as our main support-function method for compatibility with the conditional mean independence assumption of our model; the Gafarov approach provides a useful theoretical alternative for the polygon case.

4. SIMULATIONS

Various data generating processes (DGPs) are simulated to implement the estimation method described in the previous section and compare it with point-estimation obtained when using the midpoint of the interval as the true value of sales. The assumed DGPs are described as follows:

- The number of markets and products $(T, J_t) \in \{(100, 5), (1, 500)\}$
- Market size M_t is chosen such that $\min_j S_{jt} = 1$
- The bound definition rule for variables $x \in \{S_{1t}, \dots, S_{J_t}, M_t\}$:

$$x_L = \max_{(a_0, a_1) \in (A_0, A_1)} a_0 \times 10^{a_1} \leq x \quad \text{and} \quad x_U = \min_{(a_0, a_1) \in (A_0, A_1)} a_0 \times 10^{a_1} > x$$

- $A_0 = \{1\}$ and $A_1 = \mathbb{N} - \{0, 1, 2, 3\}$ to get larger intervals
- $A_0 = \{1, 2, 5\}$ and $A_1 = \mathbb{N}$ to get smaller intervals
- $X_{jt} = (1, P_{jt})'$
- true parameter $\theta_0 = (\beta_0, \beta_p) = (-7, -1.5)$
- Two instruments $Z_{jt} = (Z_{1jt}, Z_{2jt})$ generated from *Bernouilli*(0.5)
- The price is generated from a distribution that differs across DGPs according to the level of skewness of the distribution (see Figures 5, 6, 7) :
 - Low skewness or Small tail: $P_{jt} = |1 + Z_{1jt} - Z_{2jt} + \mu_{jt}| + 0.1 + \gamma$
 - Medium skewness or Medium tail: $P_{jt} = (1 + Z_{1jt} - Z_{2jt} + \mu_{jt})^2 + 0.1 + \gamma$
 - High skewness or Big tail: $P_{jt} = |1 + Z_{1jt} - Z_{2jt} + \mu_{jt}|^3 + 0.1 + \gamma$
 where $\gamma = 5$ is a constant price shifter
- $\xi_{jt} = \mu_{jt} + \nu_{jt}$
- $\mu_{jt} = \max\{-1, \min\{\eta_{jt}, 1\}\}$; where η_{jt} is a mixture of 5 normal distributions with the following vectors of mixture probabilities (0.15, 0.3, 0.2, 0.3, 0.15), means $(-0.50, -0.25, 0.00, 0.25, 0.50)$ and standard deviations (1, 1, 1, 1, 1).
- ν_{jt} is from *Uniform*(-1, 1)

The above DGPs mainly differ in the number of products, the price distribution and the range of bounds for sales. The underlying correlation between simulated prices P_{jt} , instruments Z_{jt} and demand shocks ξ_{jt} from those DGPs is reported in Table 8 in the Appendix.

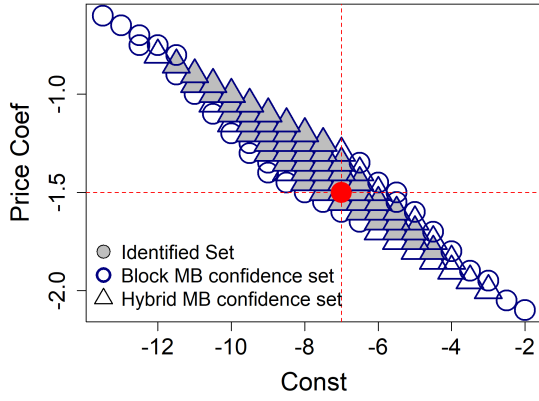
Simulation results on the CCK inference. For some DGPs, the identified set and 95%-confidence regions from a single simulated data sample are computed and presented in Figures 1. We focus on the DGPs with medium tail in the price distribution. More details and more DGPs are represented in Figures 5, 6 and 7. Besides the confidence region computed using the hybrid multiplier bootstrap (MB), I also report confidence regions obtained with the block multiplier bootstrap. The latter is less conservative and is particularly convenient in the presence of potential dependence in the data.

The figures indicate that datasets generated from DGPs with wider sales intervals tend to produce wider identified sets. Moreover, simulations with a single market and many products yield narrower confidence regions. This is driven by differences in the distribution of the implied mean-utility intervals across DGPs. In particular, when the DGP features only 5 products, the upper bounds of the mean utilities tend to dominate those generated under the 500-product specification. This occurs because, with a larger set of products, the upper bounds on sales are higher and, by Assumption 3, the corresponding lower bounds on market size are also larger. Since market size is larger in the many-products setting, the implied lower bounds on outside-good sales are higher. This, in turn, leads to smaller upper bounds for mean utilities. The larger upper bounds on sales in DGPs with 500 products are also reflected in the shape of the sales distribution. In simulations with a single market and many products, the distribution of sales is right-skewed, with a greater mass on higher values. This difference in skewness arises from the sampling design. In each simulated dataset, market size is determined so that the product with the smallest share sells exactly one unit. Consequently, when the number of products is large, the minimum share becomes very small, which implies a large market size. Sales are therefore scaled upward, generating a wider spread of sales above one and making large sales realizations more likely.

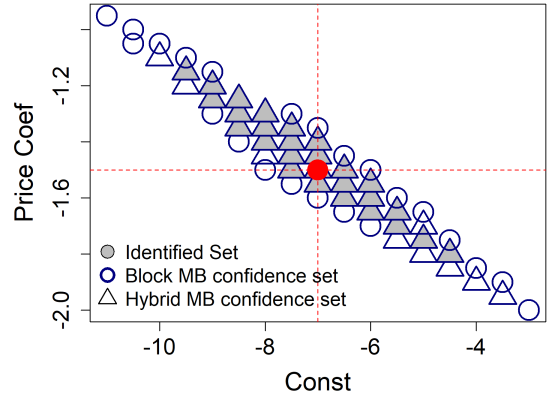
Simulation results demonstrate the presence of midpoint bias when point estimation is performed using the midpoint of interval-valued sales. The magnitude of this bias depends on the data-generating process (DGP). Based on 500 Monte Carlo simulations for each DGP, we observe systematic bias in both the price coefficient and the intercept in certain DGPs when the midpoint of the interval is treated as the true sales value (see Table 1). This occurs in the DGP with smaller intervals and few products, as well as in the DGP with larger intervals and 500 products, with the bias being more pronounced in the latter case. In these two DGPs, using the midpoint leads to a 0% probability that the confidence intervals for the

Figure 1: CCK Simulations — Identified set and 95%-confidence regions

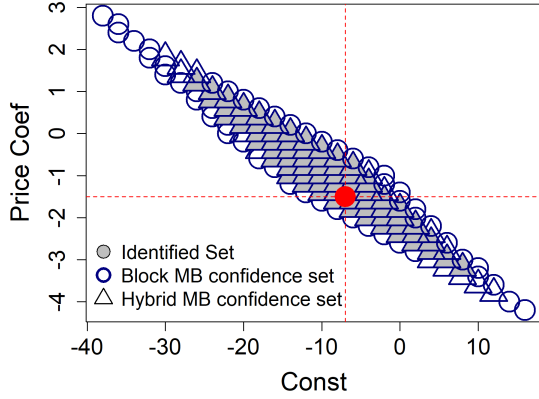
(a) Smaller intervals; Medium tail in price distribution; 100 markets; 5 products



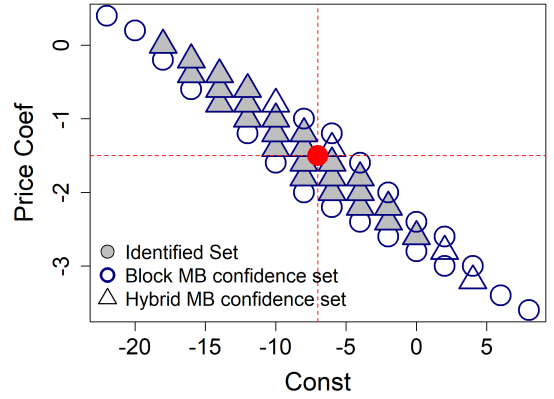
(b) Smaller intervals; Medium tail in price distribution; 1 market; 500 products



(c) Larger intervals; Medium tail in price distribution; 100 markets; 5 products



(d) Larger intervals; Medium tail in price distribution; 1 market; 500 products



Notes The point in red colour represents the true parameter $\theta_0 = (-7, -1.5)$. Parameter values from the grid that are in the identified set - in grey color - are the ones that solve the linear system defined by moment inequalities in 7. The population moments are approximated by empirical moments computed using a large sample of 50000 observations.

Table 1: CCK Simulations — 95 %-Confidence Intervals — Averages across 500 simulations

	2SLS with Interval				CCK						True Values (θ_0)
	Midpoint ^(a)				(Hybrid MB)			(Block MB)			
	Coef	Lower	Upper	C.R.	Lower	Upper	C.R.	Lower	Upper	C.R.	
Smaller intervals; Medium tail in price distribution; 100 markets; 5 products											
Const	-8.082	-8.517	-7.647	0.008	-12.307	-2.982	0.974	-13.353	-1.801	1	-7.0
Price Coef	-1.291	-1.350	-1.231	0.000	-2.031	-0.744	NA	-2.162	-0.609	NA	-1.5
Smaller intervals; Medium tail in price distribution; 1 market; 500 products											
Const	-7.145	-7.543	-6.746	0.848	-10.067	-3.341	0.948	-10.913	-2.347	0.994	-7.0
Price Coef	-1.461	-1.515	-1.406	0.670	-1.988	-1.091	NA	-2.102	-0.968	NA	-1.5
Larger intervals; Medium tail in price distribution; 100 markets; 5 products											
Const	-8.032	-9.186	-6.878	0.558	-31.780	11.876	0.96	-38.576	17.068	1	-7.0
Price Coef	-0.751	-0.909	-0.592	0.000	-3.858	2.078	NA	-4.451	2.884	NA	-1.5
Larger intervals; Medium tail in price distribution; 1 market; 500 products											
Const	-11.726	-12.261	-11.191	0	-18.684	4.492	0.984	-24.388	10.164	0.998	-7.0
Price Coef	-0.860	-0.933	-0.787	0	-3.292	0.069	NA	-3.966	0.819	NA	-1.5

(a) $\delta_{jt} = (\text{Log}(S_{jtL} + S_{jtU})) - (\text{Log}(S_{otL} + S_{otU}))$

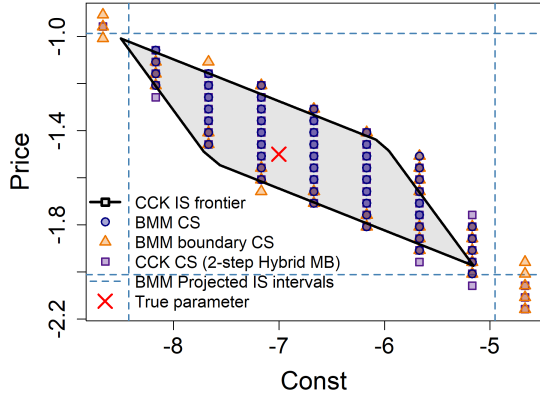
(b) For the 2SLS estimates using the midpoint of interval as the true value for sales, this is the % of simulations with the confidence intervals covering the true parameter value: the coverage rate is provided for each parameter coordinate. For the CCK estimates, this is the minimum % of simulations with the confidence region covering $\theta \in \Theta_I$: the coverage rate is computed pointwise for each $\theta \in \Theta_I$ and then minimized across parameters from Θ_I .

price coefficient and the intercept cover their true values. In contrast, confidence regions constructed using the CCK tests for moment inequalities achieve coverage probabilities close to one for all parameters in the identified set. With hybrid multiplier bootstrap (MB) critical values, coverage probabilities are bounded below by 0.974 and 0.984. Using the less conservative block MB critical values, the minimum coverage rate reaches 100%. Overall, the price coefficient tends to be underestimated across DGPs. Despite being less conservative, the confidence intervals projected from the CCK confidence region exhibit good coverage of values in the identified set, including the true parameter.

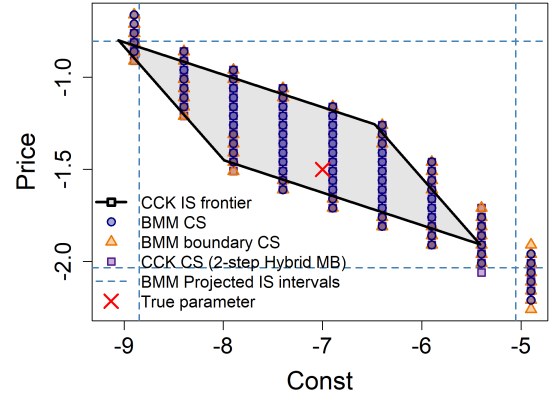
Additional simulation results on the BMM inference. We complement the CCK results with the BMM support-function-based inference of Section 3.2.2, using a new value for the constant price shifter, set to $\gamma = 1$. We characterize the identified set $\Theta_{\mathcal{G}}(P)$ in two complementary ways: (i) as the polygon obtained by the linear moment inequalities underlying the CCK inference; and (ii) as the envelope of the BMM frontier points, obtained by sampling 1,000 directions q on the unit sphere and computing the corresponding support-function maximizers. Both representations are evaluated by estimates on a single large simulated dataset ($n = 50,000$); the projected identified interval for each coordinate l is likewise read off this large sample as $[-\hat{\delta}_n^*(-e_l | \Theta_{\mathcal{G}}(P)), \hat{\delta}_n^*(+e_l | \Theta_{\mathcal{G}}(P))]$. By contrast, the confidence sets, projected confidence intervals, and coverage rates reported below are computed on small samples ($n = 500$) and averaged over 500 Monte Carlo simulations, in order to study their bias and coverage. The CCMS smoothing parameter τ_n is set to a small fraction of

Figure 2: BMM simulations — Identified set and 95%-confidence sets

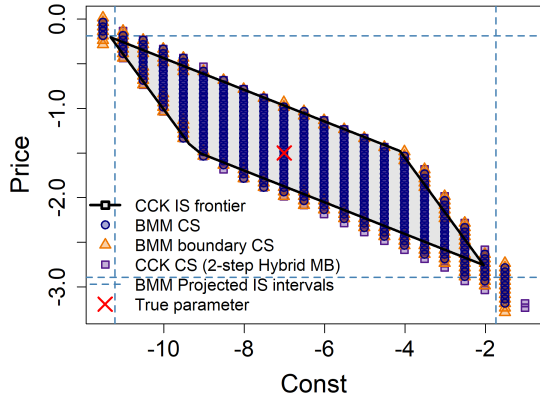
(a) Smaller intervals; 1 market; 500 products



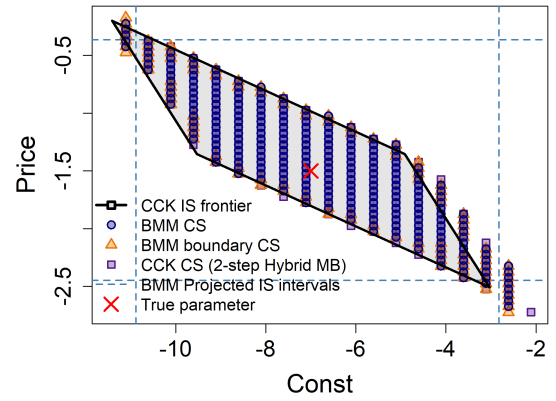
(b) Smaller intervals; 100 markets; 5 products



(c) Larger intervals; 1 market; 500 products



(d) Larger intervals; 100 markets; 5 products



Notes Grid points not rejected at the 5% level by each test, in the (Const, Price) plane for a single simulated dataset. Circles: BMM confidence set (Test 1, $H_0 : \theta_0 \in \Theta_G(P)$). Triangles: BMM boundary confidence set (Test 2, $H_0 : \theta_0 \in \partial\Theta_G(P)$). Squares: CCK confidence set (two-step hybrid MB). The grey region is the grid-approximated identified set, the solid black curve is the CCK identified-set frontier, and the dashed blue lines are the BMM projected identified intervals. The red cross is the true parameter $\theta_0 = (-7, -1.5)$.

$E[|\delta_{jtU}| + |\delta_{jtL}| + |\delta_{jtU} - \delta_{jtL}|]$ to address the mass points carried by the Bernoulli instruments Z_{1jt}, Z_{2jt} . The inner and outer perturbations are both set to $a_n^{\text{inner}} = a_n^{\text{outer}} = 0.1 \cdot n^{-1/3}$, as in Bontemps et al., 2012, Section 5.

The identified set traced by the BMM frontier points closely matches the polygon cut out by the CCK moment inequalities (Figure 8). Two practical observations about the BMM inference are worth noting.

First, the studentized test statistic $T_n^{BMM}(\theta_0)$ can be numerically unstable near the corners of $\Theta_{\mathcal{G}}(P)$, for two distinct reasons. *(a) Near-kinks.* At a corner the empirical support function is nearly non-differentiable: the maximizing direction q is almost set-valued, so the selected direction — and the variance $\hat{V}_q$ attached to it — changes abruptly as θ_0 crosses the corner. Because the instruments are discrete, the kink directions are precisely those where the direction coefficient $z_{jtq} = \frac{1}{J}q'\Omega^{-1}Z_{jt}$ vanishes for a non-negligible fraction of observations; those observations drop out of the moment $z_{jtq}\varepsilon_{jtq}$, so the effective sample behind $\hat{V}_q$ thins out and $\hat{V}_q$ becomes small and erratic. As the support gap $\hat{\delta}_n^*(q) - q'\theta_0$ is itself near zero at the corner, the studentized ratio $T_n^{BMM} = \sqrt{n}(\cdot)/\sqrt{\hat{V}_q}$ is then highly sensitive there. *(b) A high first-stage condition number.* Independently of any kink, $\hat{V}_q$ inflates when the cross-moment matrix $\Omega = E[Z_{jt}X'_{jt}]$ becomes poorly conditioned, since the variance is built from Ω^{-1} through $z_{tq} = \frac{1}{J}Z_t\Omega^{-1}q$. In our designs the conditioning is governed by the level of the price constant shifter γ : with a large constant shifter, $\text{Var}(\widehat{\text{price}})$ is small relative to its mean, so the intercept and price are nearly collinear after the instrument projection, the condition number of Ω rises, Ω^{-1} has large entries, and $\hat{V}_q$ inflates in the Const direction. Figure 9 in the Appendix makes this concrete for the larger-interval, 100-market design: at $\gamma = 10$ the first stage carries a high condition number, the estimated variance $\hat{V}_q$ and the test statistic T_n^{BMM} blow up along the Const direction, and the confidence set has big blobs at the corners of the lens; reducing the shifter to $\gamma = 1$ brings the condition number down sharply and these issues disappear. The CCMS smoothing targets the discreteness in *(a)* and does not address *(b)*, the constant–price collinearity, best controlled by reducing the shifter γ (or centering price). Weak instruments would produce the same variance inflation through a different channel, though not in our simulation design.

Second, the projected confidence intervals from BMM (computed from the $\pm e_l$ directions only) are systematically tighter than the projection of the full BMM joint confidence region. This is consistent with the standard observation in the partial-identification literature: pointwise marginal inference is less conservative than the projection of a joint confidence region (Kaido et al., 2019). In practice, the BMM projected intervals are the most useful summary when the primary interest is in individual coefficients (e.g. the price elasticity), while the CCK joint confidence region remains preferable when the interest is in the full vector θ .

Table 2: BMM Simulations — Bounds of the projected identified intervals (with 95% CIs)

	Grid IS ^(a)		Lower bound $\hat{\theta}_j^L$			Upper bound $\hat{\theta}_j^U$		
	Lower	Upper	BMM Est.	CI low	CI up	BMM Est.	CI low	CI up
Smaller intervals; 1 market; 500 products								
Const	-8.17	-5.67	-8.49	-8.77	-8.21	-5.14	-5.39	-4.89
Price	-1.86	-1.11	-1.98	-2.03	-1.93	-1.01	-1.11	-0.91
Smaller intervals; 100 markets; 5 products								
Const	-8.90	-5.90	-9.07	-9.37	-8.77	-5.38	-5.68	-5.08
Price	-1.81	-0.86	-1.92	-2.00	-1.85	-0.79	-0.90	-0.68
Larger intervals; 1 market; 500 products								
Const	-11.01	-2.51	-11.36	-11.81	-10.91	-2.07	-2.45	-1.68
Price	-2.63	-0.28	-2.76	-2.91	-2.61	-0.20	-0.40	0.01
Larger intervals; 100 markets; 5 products								
Const	-11.11	-3.11	-11.43	-11.89	-10.97	-2.98	-3.39	-2.57
Price	-2.47	-0.27	-2.53	-2.67	-2.38	-0.19	-0.39	0.00

(a) Grid-approximated identified interval: the range of grid points satisfying the moment inequalities from a large sample (size 50000), shown as a reference. $\hat{\theta}_j^U = \hat{\delta}_n^*(e_l | \Theta_G(P))$ and $\hat{\theta}_j^L = -\hat{\delta}_n^*(-e_l | \Theta_G(P))$, with 95% CIs from Bontemps et al., 2012 (Proposition 9); instruments are smoothed following Chandrasekhar et al., 2012. All entries are averages over 500 Monte Carlo replications for four medium tail DGPs. For all these four DGPs, the estimated identified interval contains $\theta_0 = (-7, -1.5)$ in every replication.

Table 3: BMM Simulations — projected 95% confidence intervals

	2SLS Midpoint ^(a)			CCK (Hybrid MB)		BMM		BMM boundary	
	Coef	Lower	Upper	Lower	Upper	Lower	Upper	Lower	Upper
Smaller intervals; 1 market; 500 products									
Const	-6.92	-7.11	-6.73	-8.95	-4.51	-8.46	-5.16	-8.51	-5.12
Price	-1.46	-1.52	-1.41	-2.23	-0.84	-2.03	-0.98	-2.06	-0.96
Smaller intervals; 100 markets; 5 products									
Const	-7.23	-7.44	-7.02	-9.75	-4.68	-9.06	-5.38	-9.11	-5.33
Price	-1.29	-1.36	-1.23	-2.21	-0.54	-2.00	-0.75	-2.03	-0.72
Larger intervals; 1 market; 500 products									
Const	-6.86	-7.07	-6.65	-12.94	-0.25	-11.46	-1.91	-11.51	-1.84
Price	-1.48	-1.54	-1.42	-3.45	0.41	-2.96	-0.06	-3.03	-0.01
Larger intervals; 100 markets; 5 products									
Const	-7.22	-7.48	-6.95	-12.81	-1.48	-11.54	-2.83	-11.60	-2.76
Price	-1.36	-1.43	-1.28	-3.14	0.35	-2.72	-0.05	-2.79	0.01

Lower/Upper are the projected confidence intervals (ranges of accepted grid points), averaged over 500 Monte Carlo replications per design (medium tail).

(a) Using the interval midpoint as the true sales value; “Coef” is the 2SLS point estimate. Coverage rates for all methods are reported separately in Table 4.

Table 4: BMM Simulations — Monte Carlo coverage rates (medium tail; 500 replications)

	Joint θ		Const		Price Coef	
	$Cov(\theta_0)$	$\min_{\Theta_G} Cov(\theta)$	$Cov(\theta_0)$	$\min_{\Theta_G} Cov(\theta)$	$Cov(\theta_0)$	$\min_{\Theta_G} Cov(\theta)$
Smaller intervals; 1 market; 500 products						
2SLS Midpoint	0.516	0.000	0.806	0.000	0.684	0.000
CCK (Hybrid MB)	1.000	0.910	1.000	1.000	1.000	1.000
BMM	1.000	0.914	1.000	1.000	1.000	1.000
BMM boundary	0.002	0.000	1.000	1.000	1.000	0.968
Smaller intervals; 100 markets; 5 products						
2SLS Midpoint	0.000	0.000	0.428	0.000	0.000	0.000
CCK (Hybrid MB)	1.000	0.978	1.000	1.000	1.000	1.000
BMM	1.000	0.926	1.000	0.998	1.000	0.992
BMM boundary	0.000	0.000	1.000	1.000	1.000	0.996
Larger intervals; 1 market; 500 products						
2SLS Midpoint	0.560	0.000	0.662	0.000	0.884	0.000
CCK (Hybrid MB)	1.000	0.858	1.000	1.000	1.000	0.998
BMM	1.000	0.824	1.000	0.998	1.000	0.988
BMM boundary	0.000	0.000	1.000	1.000	1.000	0.996
Larger intervals; 100 markets; 5 products						
2SLS Midpoint	0.038	0.000	0.636	0.000	0.038	0.000
CCK (Hybrid MB)	1.000	0.976	1.000	1.000	1.000	0.998
BMM	1.000	0.940	1.000	0.990	1.000	0.982
BMM boundary	0.000	0.000	1.000	0.994	1.000	0.988

Coverage rates over 500 Monte Carlo replications per design, with focus on medium tail DGPs. $Cov(\theta_0)$ is the fraction of replications covering the true parameter: the joint confidence set covering θ_0 (“Joint”), or the projected interval of the coordinate covering its true value (“Const”, “Price”). $\min_{\Theta_G} Cov(\theta)$ is the minimum of these coverage rates over the grid-approximated identified set: over the two-dimensional set for “Joint”, and over the projected identified interval of the coordinate for “Const” and “Price”. The 2SLS rows use the midpoint 2SLS confidence interval. “BMM boundary” inverts the two-sided test $H_0 : \theta_0 \in \partial\Theta_G(P)$ and so has zero joint coverage of interior points such as θ_0 . Coverage is evaluated with no grid slack (a point counts as covered only if the interval strictly contains it).

Figures 2 and 8 and Tables 2–4 make this concrete for four medium-tail DGPs: a narrow identified set (smaller sales intervals) versus a wide one (larger sales intervals), each simulated either with a single market of 500 products or with 100 markets of 5 products. Each design is run over 500 Monte Carlo replications. In Figure 8, the BMM boundary points sampled over the unit sphere fall on top of the CCK identified-set frontier and recover the same diagonal lens in every design, narrow for the smaller-interval DGPs and much wider for the larger ones. Table 2 reports the corresponding support-function estimates: the BMM estimated bounds of the projected identified set are consistent with the grid-approximated identified interval, and for all four DGPs the estimated identified interval contains the true parameter $\theta_0 = (-7, -1.5)$ in every replication. Figure 2 and Table 3 compare the resulting confidence sets: the BMM (Test 1) and CCK (two-step hybrid MB) sets both fill the lens, with the BMM set the tighter of the two, while the BMM boundary set (Test 2) concentrates on the estimated frontier $\partial\Theta_G(P)$.

Table 4 collects the Monte Carlo coverage rates under two notions: coverage of the true parameter θ_0 , and the minimum coverage over the (grid-approximated) identified set — the uniform criterion relevant for set inference. The joint BMM and CCK confidence regions cover θ_0 in every replication, and their minimum coverage over the identified set ranges from 0.82 to 0.94 (BMM) and 0.86 to 0.98 (CCK) across the four designs; these fall somewhat short of the nominal 0.95 at the most demanding frontier points — most visibly in the larger-interval single-market design — reflecting the discreteness of the grid, and we expect them to tighten on a grid refined around the estimated bounds, as in the application. By contrast, the 2SLS midpoint interval has zero minimum coverage over the identified set in every design: a point-identification interval cannot cover an entire set. More tellingly, with no grid slack in the coverage check, the 2SLS interval also fails to cover the true point θ_0 in a large fraction of replications — jointly 0.52 and 0.56 in the two single-market designs, and only 0.04 and 0.00 in the two 100-market designs, where the tighter midpoint interval makes the midpoint bias of Table 1 bite hardest. The BMM boundary set has essentially zero joint coverage of the interior point θ_0 by construction, since the two-sided boundary test targets $\partial\Theta_G(P)$ rather than its interior. Coordinate by coordinate, the projected BMM and CCK intervals cover the identified projection in essentially every replication (minimum coverage between 0.98 and 1.00), whereas the 2SLS projected interval again has zero minimum coverage over the identified set.

5. ILLUSTRATION WITH DATA ON STEAM APPS

5.1. Data

An industry of interest for implementing the suggested method in a demand analysis is that of digital apps, as app providers often disclose download numbers in intervals. This is the case for apps that are available on the Google’s Playstore platform. I focus on another case, that of personal computer (PC) video games that are available on the Steam platform. Data was collected monthly from the Steam API and SteamPy over 10 months on all apps available on Steam: from October 2023 to July 2024. Table 10 summarizes the main observed outcomes such as prices, interval-valued sales and user bases. Intervals for sales are derived from intervals for cumulative sales (i.e. base of users).

Markets are given by the observation periods for sales and prices, spanning November 2023 to July 2024; and sales units are consumer purchase occasions. To illustrate the suggested method in demand analysis, I focus on markets for recent video game apps from popular genres, i.e. apps from popular genres that were released between January 2023 and July 2024 inclusive. This choice is motivated by the fact that we expect lower variations in sales from old apps as they have reached the last phase in their life cycle. Only very few ones from these old video game apps are still able to attract new users. Only apps from well-known genres are considered in consumer-facing options. These popular genres are the followings: action, adventure, role-playing-game, simulation, sports, racing and education. As selection issues might arise from the heterogeneity in the product launch period and in the product deprecation period, the consumer choice problem is restricted to products that were released during the first quarter of 2023 and that were still available in July 2024. After the previous selection of products to consider, the number of remaining apps ranges from 3417 in October 2023 to 7070 in July 2024. Of these few thousand selected apps, 90 belong to the action-adventure genre and reached at least once the threshold of 20K users between 03-2023 and 07-2024. These 90 apps are the substitutable products that I consider in the consumer choice set. The other apps that are not from the action-adventure genre or never reached 20K users are pooled in the outside option. This focus on the 90 popular apps is driven by the quest for enough variability in (interval-valued) sales and user bases. Figure 11 is indicative of the much concentrated distribution of (interval-valued) user base when we consider all apps available on Steam: As of October 2023, more than half of all apps had the same interval of 0-20000. Moreover, by focusing on a quite specific group of products (recent, casual video games from action-adventure genre), the demand analysis doesn’t require us to take many product attributes into account. I am assuming that the 90 applications selected are differentiated by two observed attributes - price and online/multiplayer mode status - and by their unobserved quality.

A summary of product attributes considered in this analysis is provided in table 11. The price (P_{jt}) is an endogenous regressor in the demand equation. Exogenous variations come from the multiplayer/online mode status⁵ (MO_{jt}) and two excluded instruments for the price: the number - in log - of total apps the firm has on Steam and a dummy variable for whether or not the firm has a single app available on Steam. The online/multiplayer mode status for rival products couldn't be used as instrument because of low variations. These instruments relate to BLP instruments since they represent aggregate attributes across other products of the same firm.

The mean utility is given by:

$$\delta_{jt} = \theta_c + \theta_p P_{jt} + \theta_{om} MO_{jt} + \xi_{jt} \quad \text{and} \quad \delta_{jt} = \text{Log}(S_{jt}) - \text{Log}(S_{ot})$$

Sales quantities S_{jt} and S_{ot} are partially observed in intervals. The derived intervals for mean utilities δ_{jt} have bounds that vary as shown in Figure 12. There are enough variations in the upper bounds to make inference on demand possible.

Monthly time periods are considered as markets. Market units are purchase occasions and the sales quantity of a product is the number of purchase occasions in which consumers chose that product. Therefore, given the details previously given on markets, the market size is the total purchase occasions on Steam that led to actual purchases of popular genre apps, published between January 2023 and July 2024. Purchase occasions are assumed to be independent in their respective demand shocks, in line with assumption 1. As already mentioned, I focus on recent and casual apps from the action-adventure genre that were able to gather at least 20K units downloaded from the Steam platform. Assuming that all purchase occasions result in an actual purchase on the platform, the outside option includes other recent and casual apps from big genres that were unable to gather 20K downloads by July 2024.

5.2. Confidence set estimation

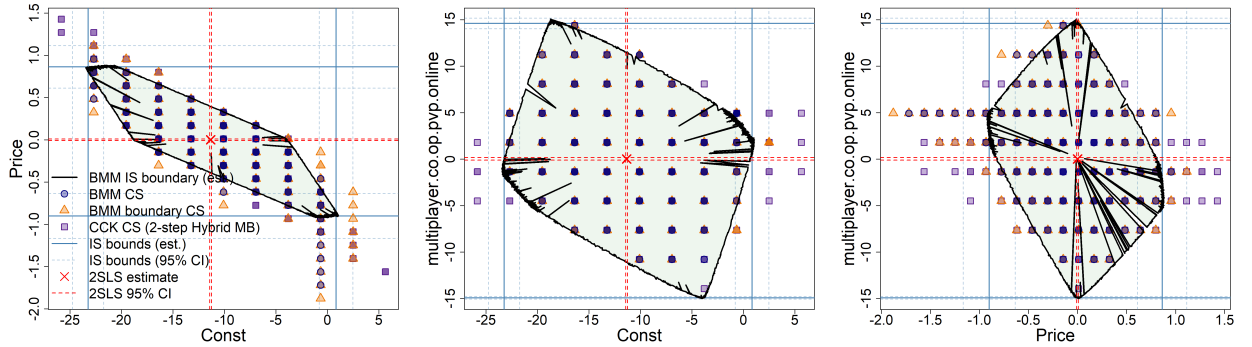
I apply the proposed approach to build confidence sets for the parameters of demand for recent casual video games from the action-adventure genre: the sensitivity to price θ_p , the sensitivity to the multiplayer/online option θ_{om} , and the average taste for unobserved attributes of the considered products relative to the outside good (the intercept θ_c). I implement both inference strategies described in Section 3.2.2 the CCK max-statistic test (two-step hybrid multiplier bootstrap) and the BMM support function-based test. Since the two excluded instruments and the multiplayer/online dummy are discrete, the empirical identified set is a

⁵More precisely, this is a tag variable indicating to users that they have access to at least one of the following features if they get the product: multiplayer, person-versus-person (pvp), co-op and online gameplay

polygon with exposed faces; I therefore smooth the instruments following Chandrasekhar et al., 2012 before applying the BMM test, scaling the perturbation as a small fraction of the average mean-utility interval length. The parameter grid contains 10,336 points spanning the three coordinates.

The support function evaluator also lets me trace the empirical identified set directly. Figure 13 plots its boundary in the three two-dimensional coordinate projections, obtained by sampling directions on the unit sphere and computing the corresponding frontier points (Section 3.2.2). The estimated identified set is a bounded convex body strictly contained in the parameter grid, which is itself evidence that the demand parameters are set-identified rather than unbounded. Figure 3 overlays, in the same projections, the parameter values not rejected by each test (BMM, BMM boundary, and CCK), together with the BMM identified-set boundary and the projected identified-set bounds. Figure 14 shows the same objects in the joint three-dimensional space, where the confidence sets concentrate on a thin slab cutting through the identified set.

Figure 3: Steam apps — Projections of the estimated identified set and 95%-confidence sets



Notes Grid points not rejected at the 5% level by each test, projected onto the three coordinate pairs. Circles: BMM confidence set (Test 1, $H_0 : \theta_0 \in \Theta_G(P)$). Triangles: BMM boundary confidence set (Test 2, $H_0 : \theta_0 \in \partial\Theta_G(P)$). Squares: CCK confidence set (two-step hybrid MB). The solid black curve connects a subsample of BMM-estimated identified-set boundary points sorted by polar angle around their centroid, forming an approximate frontier of the estimated identified set. Solid and dashed blue lines are the estimated projected bounds and their 95% confidence intervals.

Table 5 reports the BMM estimated bounds of the projected identified set for each coordinate, with 95% confidence intervals from the asymptotic Gaussian distribution of the support function estimator. Table 6 reports the projected confidence intervals obtained by inverting each test over the grid, alongside the 2SLS point estimates that treat the interval midpoint as the true sales value. Of the 10,336 grid points, the BMM test retains 173 (1.67%), the BMM boundary test 130 (1.26%), and the CCK test 195 (1.89%).

The results carry three messages. First, on the *price coefficient*, the estimated identified-set bounds $[-0.90, 0.86]$ straddle zero, so the price sensitivity is set-identified and the data alone cannot determine its sign. Both partial-identification confidence sets reflect this: the BMM confidence interval for θ_p is $[-1.72, 0.95]$ and the CCK interval is $[-1.56, 1.43]$, each admitting

Table 5: BMM estimated bounds of the projected identified set (95% CIs)

Coef	Grid			Lower bound $\hat{\theta}_j^L$			Upper bound $\hat{\theta}_j^U$		
	Values	Lower	Upper	Est.	CI lower	CI upper	Est.	CI lower	CI upper
Const	16	-35.27	11.93	-23.24	-24.76	-21.72	0.83	-0.80	2.47
Price	34	-2.66	2.53	-0.90	-1.17	-0.64	0.86	0.61	1.11
multiplayer/online	19	-29.68	26.97	-14.91	-15.05	-14.78	14.62	14.04	15.19

Notes For coordinate j , the upper bound is $\hat{\theta}_j^U = \hat{\delta}_n^*(e_l | \Theta_{\mathcal{G}}(P))$ and the lower bound is $\hat{\theta}_j^L = -\hat{\delta}_n^*(-e_l | \Theta_{\mathcal{G}}(P))$, with 95% confidence intervals from Bontemps et al., 2012 (Proposition 9). Instruments are smoothed following Chandrasekhar et al., 2012. The “Grid” block reports the number of grid values for each coordinate and the grid lower and upper bounds. “multiplayer/online” is the coefficient θ_{om} of the multiplayer/co-op/pvp/online dummy.

Table 6: Projected 95% confidence intervals: BMM, CCK, and 2SLS

Coef	BMM CS		BMM boundary CS		CCK CS		2SLS ^(a)		
	lower	upper	lower	upper	lower	upper	Coef	lower	upper
Const	-22.69	-0.66	-22.69	2.49	-25.83	5.64	-8.4430	-8.5742	-8.3118
Price	-1.72	0.95	-1.88	1.11	-1.56	1.43	0.0126	0.0009	0.0242
multiplayer/online	-10.80	14.38	-10.80	14.38	-13.94	14.38	0.0476	-0.1563	0.2514

(a) Using the midpoint of the interval as the true value for sales; 2SLS point estimate with confidence interval for each coordinate. CCK CS uses the two-step hybrid MB critical values. Projected confidence intervals are the ranges of accepted grid points; the grid spans $[-40, 10] \times [-2.5, 2.5] \times [-20, 20]$.

the economically sensible region of negative price sensitivity while also failing to exclude zero and small positive values. By contrast, the 2SLS estimate that treats the interval midpoint as the true sales value yields a small *positive* and statistically significant price coefficient (0.0126, CI [0.0009, 0.0242]), an implausible upward-sloping demand. This is exactly the midpoint bias documented in the simulations of Section 4: collapsing the interval to its midpoint produces precise but misleadingly signed estimates, whereas the partial-identification confidence sets recover the plausible negative region at the cost of width.

Second, on *methodological comparison*, the BMM confidence set is tighter than the CCK confidence set. It retains fewer grid points (173 versus 195) and yields narrower projected intervals for the intercept and the price coefficient (for example, $[-22.69, -0.66]$ versus $[-25.83, 5.64]$ for the constant). This is consistent with the general observation that direct marginal inference is less conservative than the projection of a joint moment-inequality confidence region (Kaido et al., 2019), and with the simulation evidence in Section 4. The BMM boundary confidence set, which inverts the two-sided test of $H_0 : \theta_0 \in \partial\Theta_{\mathcal{G}}(P)$, is narrower still on the side where the identified set is sharply estimated.

Third, on the *multiplayer/online coefficient*, the identified-set bounds $[-14.91, 14.62]$ are very wide and the confidence sets span almost the entire grid range, so the data are essentially uninformative about θ_{om} : its sign cannot be determined and zero is always included for any value of the price coefficient in the confidence set. The limited and discrete variation in the multiplayer/online attribute, together with the partial observation of sales, leaves this

coefficient effectively unidentified in practice. The projected confidence intervals for the constant lie almost entirely in the negative region under the BMM test ($[-22.69, -0.66]$), consistent with a low average taste for the inside goods relative to the large outside option. Across coordinates, the projected confidence intervals extend somewhat beyond the point estimates of the identified-set bounds but remain comparable to the bounds' own confidence intervals (Table 5); this is expected, since the confidence sets account for the sampling uncertainty of the estimated identified set rather than conditioning on its point estimate. Point observation of sales, or additional demand-side restrictions, would be required to sharpen inference on any individual coefficient.

6. A SECOND ILLUSTRATION: PUZZLE APPS ON THE GOOGLE PLAY STORE

The Google Play Store reports each app's install count only as one of a small number of coarse brackets (1+, 5+, 10+, ..., up to billions of installs), so an app's installed base is observed as an interval rather than as a number. App demand on the Play Store is therefore a natural second setting for the interval-valued approach, with the sales bounds arising directly from the reporting convention rather than from a measurement device. We focus on the puzzle category: a large set of close substitutes with wide variation in monetization (free, ad-supported, in-app purchases), a moderate quality range, and neither a subscription structure nor two-sided-market features, so a static discrete-choice demand model is appropriate.

6.1. Data

The data are a public scrape of the Google Play Store. To hold the release year fixed and avoid confounding from app age, cohort, and survivorship effects, the inside options are restricted to free puzzle apps from a single release cohort (2019); the outside option aggregates the remaining puzzle apps in the market. After dropping one app with missing rating information, the estimation sample is $n = 799$ apps in a single market. Because installs are reported in 19 ordered brackets, app j 's installs are known only to lie in $[S_{jL}, S_{jU}]$. We normalize each app's installs to a per-day flow, as a ratio between the installs observed during the data collection period, in Mid-2021, and the age of the app (in days). The outside-option sales bounds $[S_{oL}, S_{oU}]$ are obtained by aggregating the brackets of the outside apps, and the market size satisfies $M \in [M_L, M_U]$. The mean utility $\delta_j = \log s_j - \log s_0$ is then bounded by $\delta_{jL} = \log S_{jL} - \log S_{oU}$ and $\delta_{jU} = \log S_{jU} - \log S_{oL}$.

Mean utility is modeled as $\delta_j = X_j' \theta + \xi_j$ with nine regressors: a constant; five endogenous monetization and quality characteristics — an indicator for in-app purchases, an indicator for being ad-supported, an indicator for a non-zero star rating, the log rating count per

day, and an indicator for a high daily rating count — and three exogenous characteristics — the log app size, a post-release update measure, and an indicator for whether the developer publishes other puzzle apps. The endogenous characteristics are instrumented with BLP-style developer aggregates — sums, over the developer’s other puzzle and non-puzzle apps, of log size, dummy for post-release update, and dummy indicating the minimum Operating System version to Android 5 or higher — together with two non-BLP instruments that exploit the cross-platform match between a Google app and its Apple counterpart: the size gap and the age gap across platforms. The raw instruments are passed through a hypercube transformation (following D. W. Andrews and Shi, 2013) and a selection step (to ensure full rank condition is satisfied), leaving 40 instruments. Summary statistics for the estimation sample are reported in Appendix Table 12.

The cross-platform instruments are useful because they capture platform-specific variation in how the same underlying product is adapted, released, and maintained across Google Play and the Apple App Store. Such variation can affect monetization and rating outcomes on Google Play. For instance, a larger size difference across platforms may reflect differences in platform-specific technical implementation, advertising or in-app-purchase modules, or adaptation effort. Similarly, a release-age gap across platforms may reflect rollout timing and platform prioritization, which can affect the timing of user exposure and rating accumulation. Conditional on the Google Play characteristics included in the demand equation and on developer portfolio instruments, the identifying assumption is that these cross-platform gaps affect Google Play demand only through the endogenous monetization and rating variables, not directly through the Google Play demand shock ξ_j .

This exclusion restriction is credible only if the cross-platform gaps mainly reflect platform-specific implementation and rollout decisions rather than unobserved app quality that directly affects Google Play demand. For this reason, the cross-platform instruments are best interpreted as additional sources of identifying variation that complement the developer aggregates. Their role is not to instrument app size or post-release updates, which are included as controls, but to help separate the effects of monetization and rating activity from unobserved demand shocks. The raw instruments are then passed through the hypercube transformation of Andrews and Shi (2013), followed by a selection step to ensure that the full-rank condition is satisfied, leaving 40 instruments in the final specification.

6.2. Identified bound estimation

Table 7 reports the results for the Google Play Store puzzle-app application. The first block gives the 2SLS estimates obtained after replacing each install interval by its midpoint. The second and third blocks report the lower and upper bounds of the BMM projected identified set for each coefficient, together with 95% confidence intervals for these projected bounds.

Figure 4 complements the table by plotting two-dimensional projections of the estimated identified set onto each (constant, coefficient) plane.

The BMM empirical identified set is built as follows. I trace its frontier by evaluating the empirical support function over 10,000 directions on the unit sphere. The projected intervals in Table 7 are obtained from the $\pm e_l$ coordinate directions. As in the simulations, the instruments are smoothed following Chandrasekhar et al., 2012. Since the final instrument set contains more instruments than parameters, the support function is evaluated in its overidentified form using a projected-gradient optimizer, with inner perturbation $a_n = 0.1 n^{-1/3}$. The 95% confidence intervals for the projected bounds are computed using the asymptotic Gaussian distribution of the support-function estimator in Bontemps et al., 2012. The final instrument set contains 40 hypercube-transformed instruments.

The contrast between the midpoint estimates and the partial-identification results is clear. The midpoint 2SLS regression suggests a relatively sharp demand relationship. It gives positive and statistically significant coefficients for non-zero rating, log rating count per day, post-release updates, and having more puzzle apps by the same developer. These signs are natural at first sight. Apps with user ratings, more rating activity, updates, and developers with experience in the puzzle category may be expected to obtain more installs. However, these conclusions rely on treating the midpoint of each install bracket as if it were the true number of installs.

The BMM results show that the interval-valued install data support a much wider set of economic interpretations. At the estimated identified-set level, only two coefficients rule out zero: the constant and log rating count per day. The constant has an estimated identified interval $[32.83, 7.01]$, which is entirely negative. This is consistent with low average utility for each inside app relative to the outside option. The coefficient on log rating count per day has an estimated identified interval $[0.06, 3.56]$, which is positive. This is the most robust positive demand shifter in the application. It suggests that rating intensity, which captures visibility, user engagement, and accumulated attention, is strongly related to demand.

For all other coefficients, the estimated identified intervals contain zero. This includes in-app purchases, ad support, non-zero rating, high rating count per day, app size, post-release update, and the developer puzzle-portfolio indicator. Once sampling uncertainty is added, the confidence intervals around the projected bounds become even wider. In fact, the confidence intervals of the projected bounds include zero even for the constant and log rating count per day. Therefore, the BMM results should be interpreted as showing which signs are supported by the estimated identified set, while also making clear that sampling uncertainty remains important.

The midpoint 2SLS estimate lies inside the BMM identified interval for every coefficient. This means that the midpoint regression does not give a value that is impossible under the

interval data. The problem is different: it selects one point from a wide set and makes this point appear much more precise than the data justify. The partial-identification approach shows that the same interval-valued install data are compatible with many combinations of demand parameters.

Figure 4 is important for interpreting this result. The projections show that the identified set is not simply a collection of independent intervals. There are clear trade-offs between the constant and the other coefficients. The constant captures the baseline utility of inside apps relative to the outside option. Since the outside option is large in this application, a very negative constant is plausible. A constant closer to -10 is less negative and corresponds to parameter configurations where the inside apps have higher baseline utility relative to the outside option. Equivalently, within the admissible sales bounds, these configurations are consistent with the outside option being less dominant.

When the constant is closer to -10 , the frontier of the identified set tends to give positive effects to in-app purchases, non-zero rating, log rating count per day, and having more puzzle apps by the same developer. These signs have reasonable economic interpretations. In-app purchases may proxy for richer content, stronger monetization effort, or higher developer investment. A non-zero rating may signal that the app has received some attention from users. The positive coefficient on log rating count per day is especially meaningful because it remains positive across the estimated projected interval. Having more puzzle apps by the same developer may capture experience in the category, cross-promotion, or a better understanding of puzzle-game users.

In the same region of the identified set, where the constant is closer to -10 , the coefficients on ad support, log app size, and post-release updates tend to be negative. Some of these signs are economically plausible. A negative effect of ad support can reflect the direct disutility of ads or the fact that ad-supported apps may be perceived as lower quality. A negative effect of app size can reflect download frictions, storage constraints, or lower accessibility on some devices. The negative effect of post-release updates is more difficult to interpret. Updates are often expected to improve quality, but in this setting they may also proxy for apps that needed bug fixes, apps that were initially incomplete, or developer maintenance decisions that are correlated with unobserved demand shocks. Since post-release updates are treated as a control rather than instrumented as an endogenous variable, this coefficient should be interpreted cautiously.

The coefficient on high rating count per day is also not robustly signed. This is not surprising because the model already includes log rating count per day. The continuous rating-intensity variable captures most of the relationship between rating activity and demand. The high-rating-count indicator therefore acts more like an additional nonlinear component, and the interval-valued data do not identify its sign separately with precision.

Overall, the Google Play Store application reinforces the main message of the paper. The midpoint regression suggests more sign certainty than the data can support. The BMM approach gives a less sharp but more honest description of the evidence. It shows that rating intensity is the most robust positive demand shifter in this application, while the effects of monetization choices, ad support, updates, size, and developer portfolio characteristics remain weakly identified. The projections also show that the economic interpretation of these variables depends on how the model allocates demand between baseline utility and observed app characteristics. This is precisely what partial identification is useful for: it does not force the data to choose a single parameter vector when the observed sales intervals are compatible with several economically different explanations.

Table 7: Google Play puzzle apps: 2SLS (midpoint) estimates and BMM estimated bounds of the projected identified set (95% CIs)

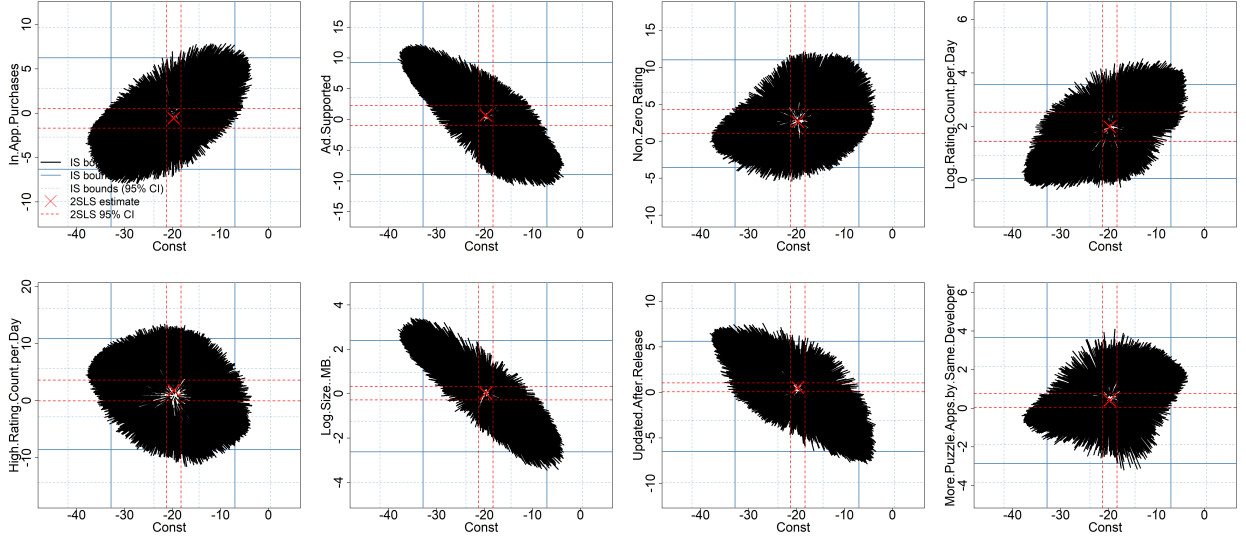
Coef	2SLS Midpoint			Lower bound $\hat{\theta}_j^L$			Upper bound $\hat{\theta}_j^U$		
	Est.	CI lower	CI upper	Est.	CI lower	CI upper	Est.	CI lower	CI upper
Const	-19.75	-21.29	-18.21	-32.83	-41.79	-23.87	-7.01	-14.47	0.46
In-app purchases	-0.59	-1.69	0.51	-6.33	-9.95	-2.71	6.24	2.82	9.66
Ad-supported	0.66	-0.96	2.28	-8.94	-13.32	-4.56	9.24	3.53	14.96
Non-zero rating	2.65	1.04	4.26	-3.56	-8.14	1.03	11.01	6.63	15.39
Log rating count per day	1.99	1.45	2.53	0.06	-0.79	0.91	3.56	1.43	5.70
High rating count per day	1.77	-0.05	3.59	-8.57	-14.31	-2.84	10.87	5.59	16.15
Log size (MB)	0.02	-0.28	0.32	-2.62	-4.00	-1.25	2.40	0.95	3.84
Updated after release	0.55	0.06	1.03	-6.50	-9.89	-3.11	5.62	2.02	9.23
More puzzle apps	0.40	0.04	0.76	-2.87	-3.84	-1.89	3.67	2.17	5.17

Notes $n = 799$ puzzle apps in a single market. For coordinate j , the upper bound is $\hat{\theta}_j^U = \hat{\delta}_n^*(e_l \mid \Theta_G(P))$ and the lower bound is $\hat{\theta}_j^L = -\hat{\delta}_n^*(-e_l \mid \Theta_G(P))$, with 95% confidence intervals from Bontemps et al., 2012 (Proposition 9). Instruments (40, hypercube-transformed) are smoothed following Chandrasekhar et al., 2012. The “2SLS Midpoint” block reports the point estimate from regressing the interval midpoint $Y_{\text{Mid},j} = \log(S_{jL} + S_{jU}) - \log(S_{oL} + S_{oU})$ on the regressors, with 95% confidence intervals.

7. CONCLUSION

The aim of this study is to estimate demand when market sales are partially observed in interval values. I propose two complementary inference strategies. The first applies the CCK max-statistic test of Chernozhukov et al., 2019 to the moment inequalities implied by partial identification under interval-valued sales. The second exploits the convex-geometric structure of the identified set: building on Bontemps et al., 2012 (BMM), I estimate the support function of the identified set and use it both to do inference on identified bounds of individual coefficients and to trace the boundary of the empirical identified set. Discreteness in the instruments (a common feature in demand estimation, where dummies and indicator-based instruments are routine) creates non-differentiability in the support function and is addressed by the smoothing technique of Chandrasekhar et al., 2012 (CCMS), which perturbs the

Figure 4: Google PlayStore puzzle apps: identified set frontier and 2SLS estimates



Notes Each panel shows the projection of the nine-dimensional estimated identified set onto the (Const, θ_j) plane. Black: BMM boundary draws tracing the frontier of $\hat{\Theta}_{\mathcal{G},n}$ (10,000 unit-sphere directions). Blue solid lines: projected identified-set bounds; faded blue dashed lines: their 95% confidence intervals. Red cross: 2SLS midpoint point estimate; red dashed lines: 2SLS 95% confidence interval. The results were quite the same when I restricted the outside goods to puzzle apps released in 2019 that exclude inside goods.

instruments with small Gaussian noise. The BMM approach has the additional advantage that the asymptotic distribution of the test statistic is explicitly Gaussian, leading to fast inference via closed-form critical values.

I compare the confidence set, the projected identified and confidence intervals from both approaches with confidence intervals that would be obtained under point identification using the interval midpoint as the true value of sales. Simulations using various DGPs provide evidence that point estimates of demand using the midpoint of the interval as the true value for sales can lead to systematic bias. The bias-free alternative, applying partial identification, gives wider but honest confidence regions; the BMM support function-based projected intervals provide a useful summary when the focus is on individual coefficients, such as the price elasticity.

I also illustrate the estimation strategy using demand data for recent and casual PC video games that belong to the action-adventure genre and that reached the threshold of 20K users by July 2024. In this illustration, the estimated confidence region allows negative values for the price coefficient, while these are ruled out by the confidence intervals computed when using the midpoint of the interval as the true value for sales. A second application uses Google PlayStore data on free puzzle apps. This application gives a richer setting, with several monetization, rating, app-characteristic, and developer-portfolio variables. The midpoint 2SLS estimates suggest that several coefficients are statistically different from zero, including non-zero rating, log rating count per day, post-release updates, and whether the

developer has other puzzle apps. However, the BMM projected identified intervals and their bounds' confidence intervals show that these sharp conclusions are not fully supported once the interval nature of installs is taken seriously. At the estimated identified-set level, only the intercept and the log rating-count-per-day coefficient rule out zero. The negative intercept is consistent with low baseline utility of inside apps relative to the outside option, while the positive coefficient on log rating count per day suggests that rating intensity is the most robust positive demand shifter in this application. For the other coefficients, the identified intervals contain zero, and the confidence intervals around the projected bounds become even wider once sampling uncertainty is added.

The Steam and Google PlayStore applications also show that partial identification is useful not only because it gives wider intervals, but because it reveals trade-offs between parameters. In particular, in the Google PlayStore application, when the constant is less negative, the identified set allows positive effects of in-app purchases, non-zero rating, log rating count per day, and developer experience in the puzzle category, while it can allow negative effects of ad support, app size, and post-release updates. Some of these signs have natural economic interpretations, while others are more difficult to interpret and may reflect remaining endogeneity, limited variation, or the weak information contained in install intervals. The main point is that the midpoint estimator selects one parameter vector from a larger set and treats it as precise, whereas the partial-identification approach shows the range of economic explanations that remain compatible with the observed data.

Overall, the results show that interval-valued sales should be treated as such. Replacing intervals with midpoints is convenient, but it can create artificial precision and misleading economic conclusions. Partial identification does not force the data to say more than they can. Instead, it separates the demand patterns that are robustly supported by the data from those that remain uncertain.

A. SUPPLEMENT ON SIMULATIONS

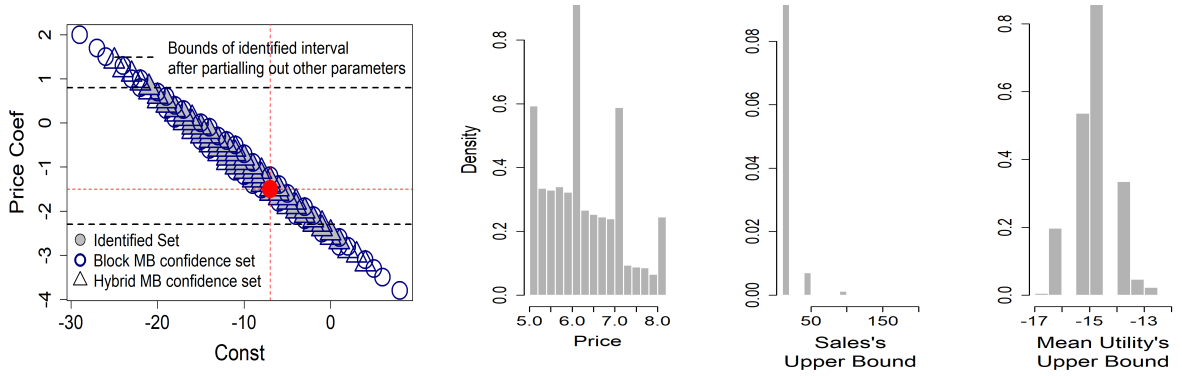
A.1. CCK Inference

Table 8: Key correlations from various DGPs used in simulations

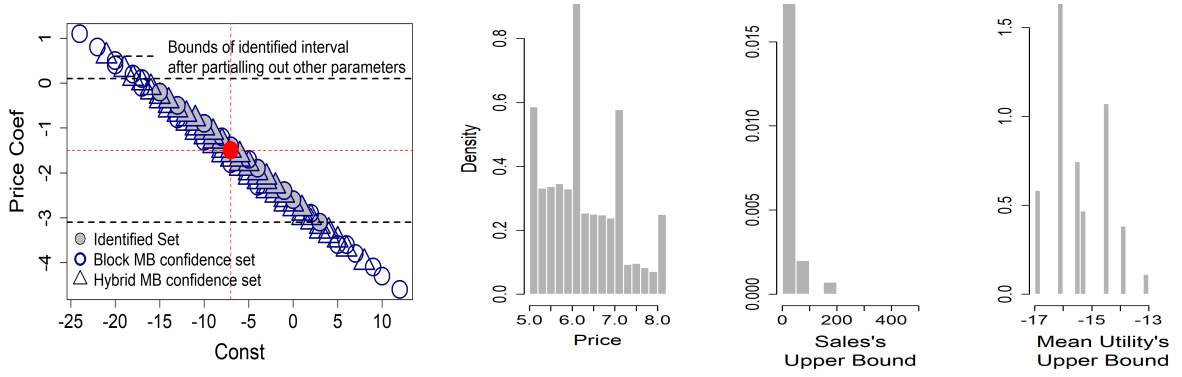
			$Cor(P_{jt}, \xi_{jt})$	$Cor(P_{jt}, Z_{1jt})$	$Cor(P_{jt}, Z_{2jt})$	$Cor(Z_{1jt}, \xi_{jt})$	$Cor(Z_{2jt}, \xi_{jt})$
Small tail in price distri- bution	Smaller intervals	T=100, J=5	0.52287	-0.40555	0.40524	-0.00041	-0.00026
		T=1, J=500	0.52468	-0.41182	0.41345	-0.00403	0.00367
	Larger intervals	T=100, J=5	0.51354	-0.40751	0.41513	0.00384	-0.00187
		T=1, J=500	0.52307	-0.41134	0.41465	0.00054	0.00293
Medium tail in price distri- bution	Smaller intervals	T=100, J=5	0.48892	-0.42161	0.42164	-0.00617	0.00476
		T=1, J=500	0.49000	-0.42011	0.41771	-0.00155	0.00669
	Larger intervals	T=100, J=5	0.47617	-0.41355	0.42188	0.01705	-0.00921
		T=1, J=500	0.48107	-0.42331	0.42424	0.00045	-0.00321
Big tail in price distri- bution	Smaller intervals	T=100, J=5	0.43546	-0.39404	0.39672	0.00292	0.00102
		T=1, J=500	0.43763	-0.39783	0.39406	-0.00089	-0.00950
	Larger intervals	T=100, J=5	0.44239	-0.40033	0.39587	-0.01136	0.00337
		T=1, J=500	0.44010	-0.39375	0.39552	0.00030	-0.00433

Figure 5: CCK Identified set, 95%-confidence regions and distributions of key variables

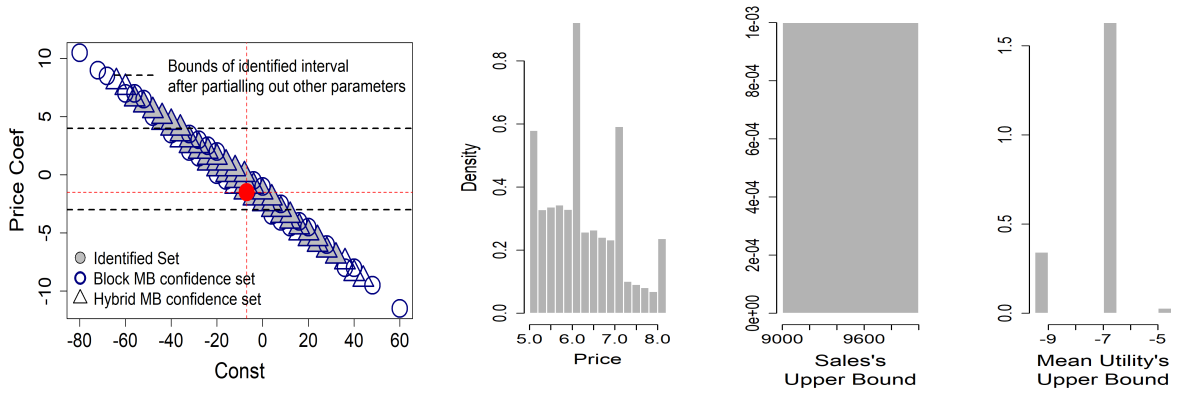
(a) Smaller intervals; Small tail in price distribution; 100 markets; 5 products



(b) Smaller intervals; Small tail in price distribution; 1 market; 500 products



(c) Larger intervals; Small tail in price distribution; 100 markets; 5 products



(d) Larger intervals; Small tail in price distribution; 1 market; 500 products

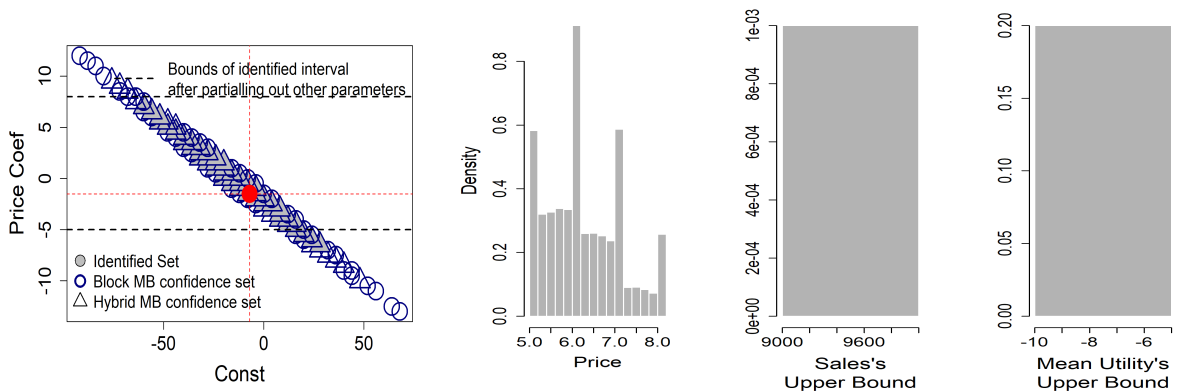
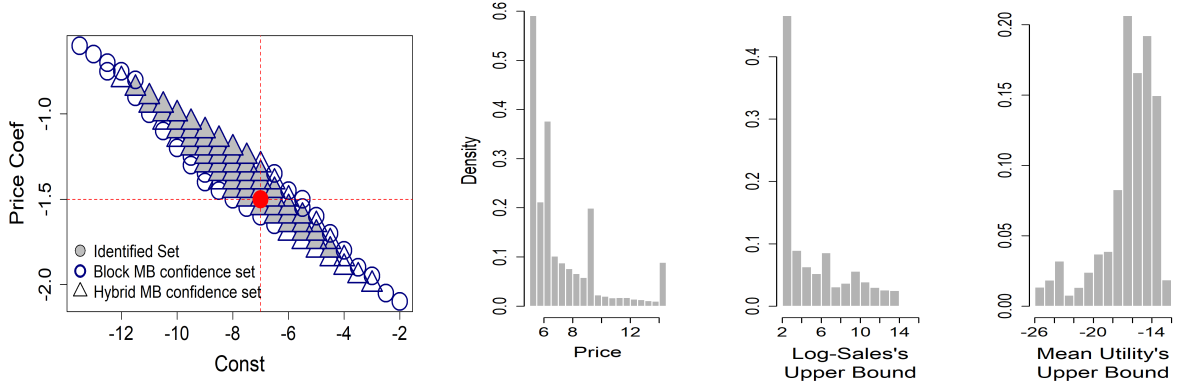
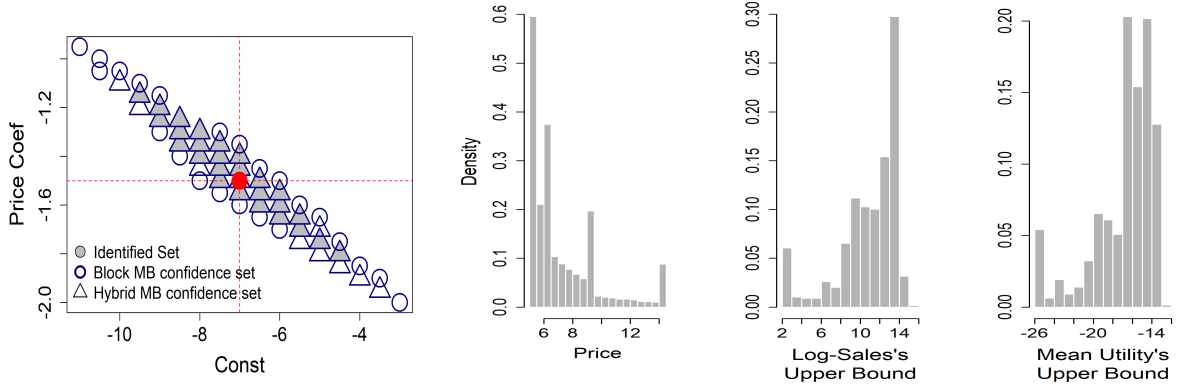


Figure 6: CCK Identified set, 95%-confidence regions and distributions of key variables

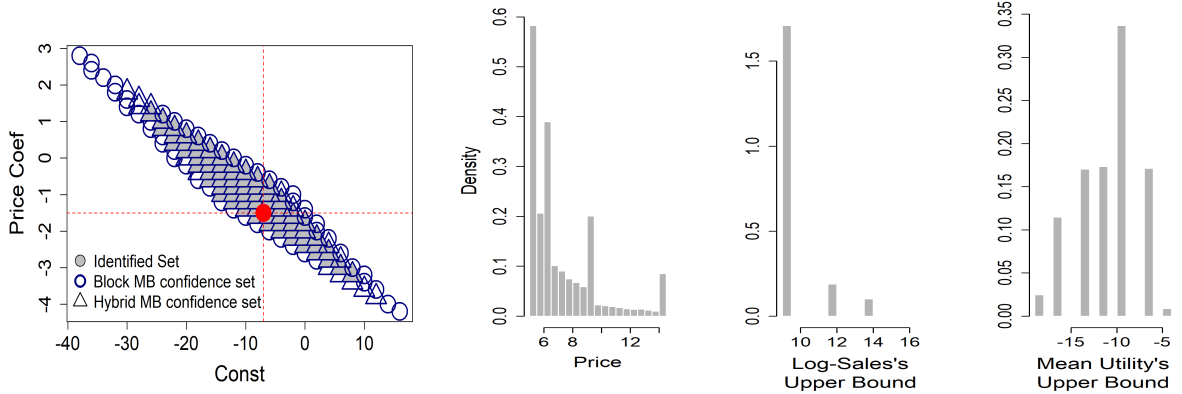
(a) Smaller intervals; Medium tail in price distribution; 100 markets; 5 products



(b) Smaller intervals; Medium tail in price distribution; 1 market; 500 products



(c) Larger intervals; Medium tail in price distribution; 100 markets; 5 products



(d) Larger intervals; Medium tail in price distribution; 1 market; 500 products

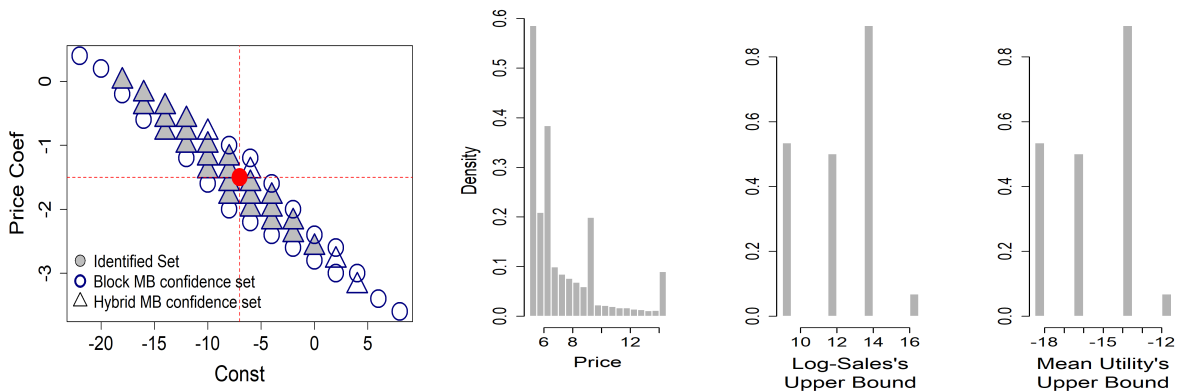
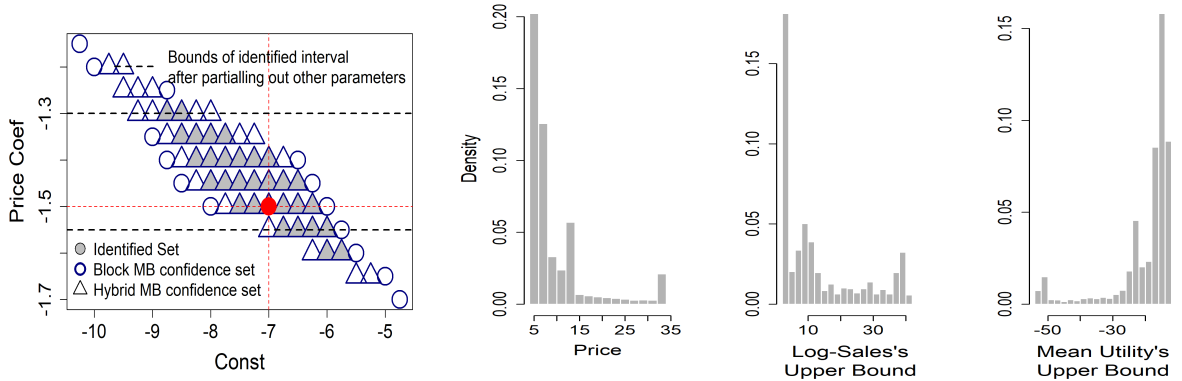
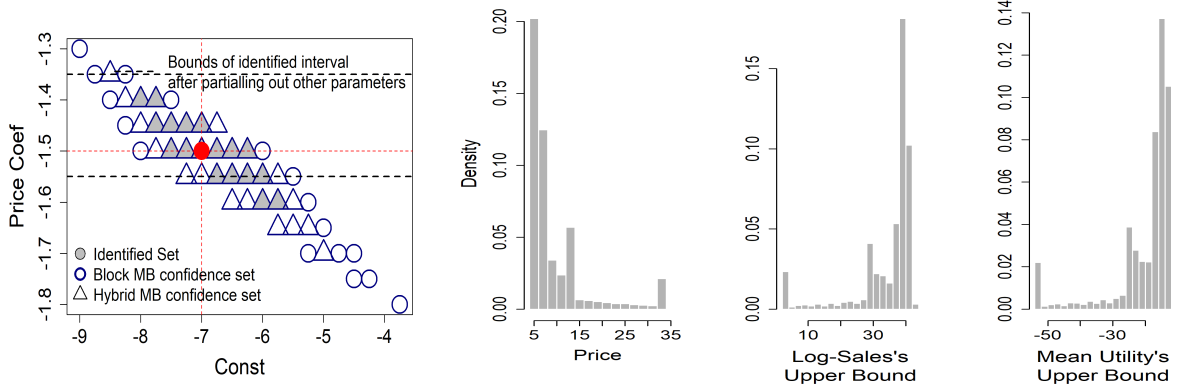


Figure 7: CCK Identified set, 95%-confidence regions and distributions of key variables

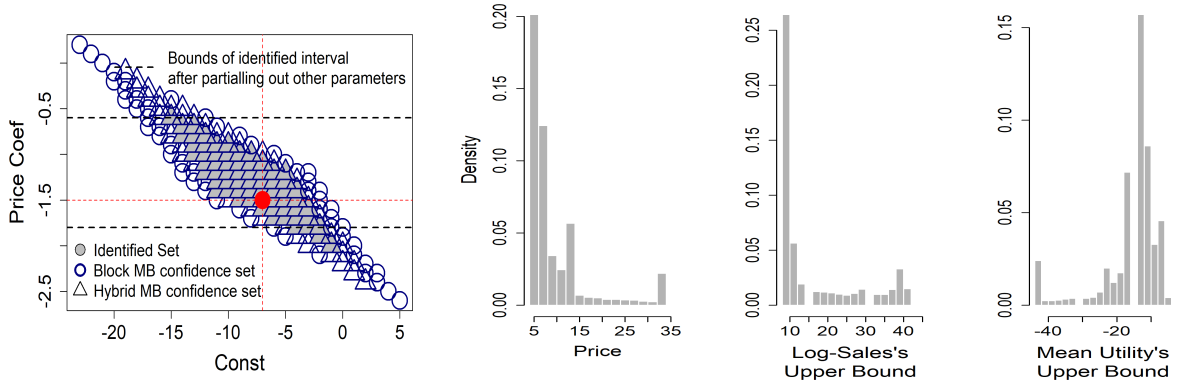
(a) Smaller intervals; Big tail in price distribution; 100 markets; 5 products



(b) Smaller intervals; Big tail in price distribution; 1 market; 500 products



(c) Larger intervals; Big tail in price distribution; 100 markets; 5 products



(d) Larger intervals; Big tail in price distribution; 1 market; 500 products

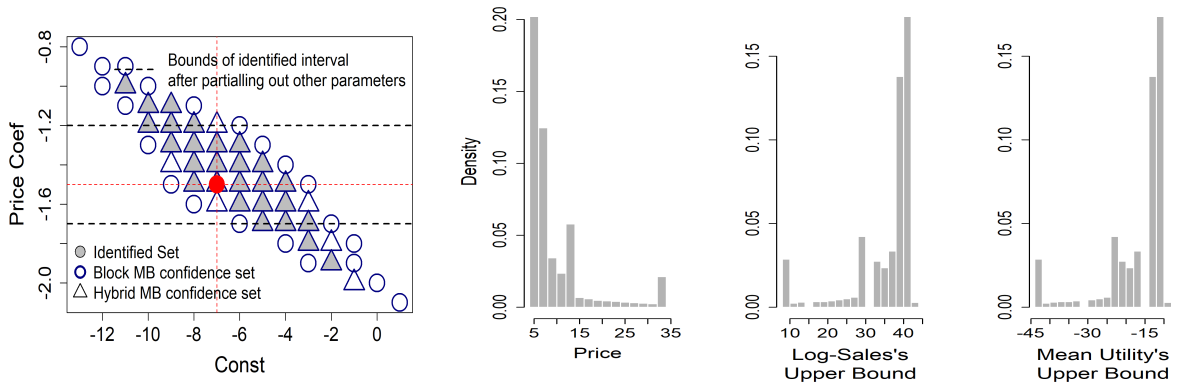


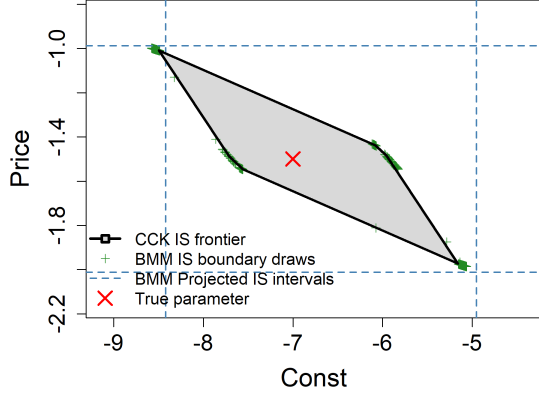
Table 9: CCK 95 %-Confidence Intervals: Averages across 500 simulations

	2SLS with Interval Midpoint ^(a)				CCK						True
	Coef	Lower	Upper	C.R.	(Hybrid MB)			(Block MB)			Values
					Lower	Upper	C.R.	Lower	Upper	C.R.	(θ_0)
Smaller intervals; Small tail in price distribution; 100 markets; 5 products											
Const	-11.192	-11.981	-10.404	0	-24.426	4.368	0.942	-27.456	7.816	1	-7.0
Price Coef	-0.731	-0.857	-0.606	0	-3.246	1.298	NA	-3.762	1.771	NA	-1.5
Smaller intervals; Small tail in price distribution; 1 market; 500 products											
Const	-8.659	-9.612	-7.707	0.148	-18.042	5.912	0.638	-21.090	9.018	0.944	-7.0
Price Coef	-1.221	-1.373	-1.070	0.094	-3.584	0.225	NA	-4.046	0.726	NA	-1.5
Larger intervals; Small tail in price distribution; 100 markets; 5 products											
Const	-8.018	-8.995	-7.041	0.45	-68.248	45.568	0.936	-84.144	60.720	1	-7.0
Price Coef	-0.264	-0.419	-0.108	0.00	-9.145	8.595	NA	-11.452	11.024	NA	-1.5
Larger intervals; Small tail in price distribution; 1 market; 500 products											
Const	-11.608	-11.608	-11.608	0	-72.472	41.752	0.982	-87.880	57.760	1	-7.0
Price Const	0.000	0.000	0.000	0	-8.818	8.974	NA	-11.279	11.316	NA	-1.5
Smaller intervals; Medium tail in price distribution; 100 markets; 5 products											
Const	-8.082	-8.517	-7.647	0.008	-12.307	-2.982	0.974	-13.353	-1.801	1	-7.0
Price Coef	-1.291	-1.350	-1.231	0.000	-2.031	-0.744	NA	-2.162	-0.609	NA	-1.5
Smaller intervals; Medium tail in price distribution; 1 market; 500 products											
Const	-7.145	-7.543	-6.746	0.848	-10.067	-3.341	0.948	-10.913	-2.347	0.994	-7.0
Price Coef	-1.461	-1.515	-1.406	0.670	-1.988	-1.091	NA	-2.102	-0.968	NA	-1.5
Larger intervals; Medium tail in price distribution; 100 markets; 5 products											
Const	-8.032	-9.186	-6.878	0.558	-31.780	11.876	0.96	-38.576	17.068	1	-7.0
Price Coef	-0.751	-0.909	-0.592	0.000	-3.858	2.078	NA	-4.451	2.884	NA	-1.5
Larger intervals; Medium tail in price distribution; 1 market; 500 products											
Const	-11.726	-12.261	-11.191	0	-18.684	4.492	0.984	-24.388	10.164	0.998	-7.0
Price Coef	-0.860	-0.933	-0.787	0	-3.292	0.069	NA	-3.966	0.819	NA	-1.5
Smaller intervals; Big tail in price distribution; 100 markets; 5 products											
Const	-7.289	-7.522	-7.055	0.3	-9.444	-5.016	0.972	-10.082	-4.20	0.998	-7.0
Price Coef	-1.430	-1.452	-1.407	0.0	-1.684	-1.234	NA	-1.742	-1.18	NA	-1.5
Smaller intervals; Big tail in price distribution; 1 market; 500 products											
Const	-6.915	-7.121	-6.708	0.812	-8.534	-4.975	0.966	-8.915	-4.282	0.994	-7.0
Price Coef	-1.488	-1.508	-1.468	0.748	-1.687	-1.346	NA	-1.736	-1.309	NA	-1.5
Larger intervals; Big tail in price distribution; 100 markets; 5 products											
Const	-7.008	-7.685	-6.332	0.858	-17.076	0.990	0.926	-20.544	3.756	1	-7.0
Price Coef	-1.200	-1.267	-1.134	0.000	-2.226	-0.364	NA	-2.427	-0.094	NA	-1.5
Larger intervals; Big tail in price distribution; 1 market; 500 products											
Const	-7.008	-7.320	-6.697	0.916	-11.150	-1.110	0.982	-13.514	1.636	1	-7.0
Price Coef	-1.360	-1.391	-1.329	0.000	-1.995	-0.982	NA	-2.207	-0.750	NA	-1.5

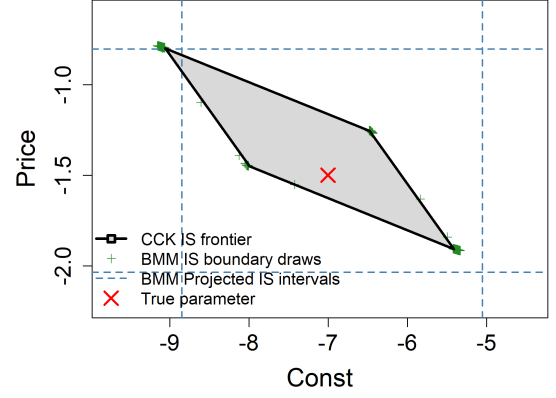
A.2. BMM vs CCK inference

Figure 8: Estimated identified set in simulations: BMM boundary draws

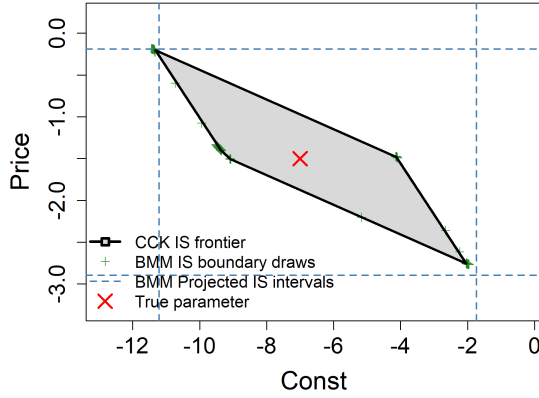
(a) Smaller intervals; 1 market; 500 products



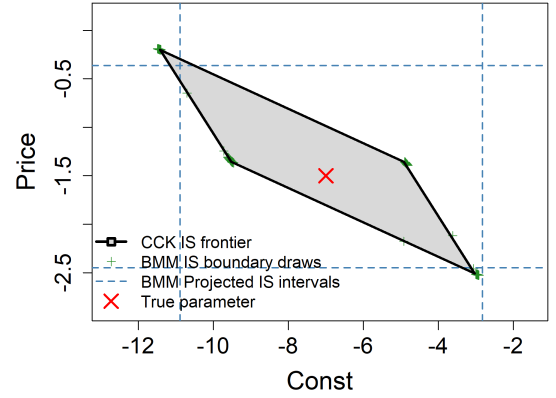
(b) Smaller intervals; 100 markets; 5 products



(c) Larger intervals; 1 market; 500 products

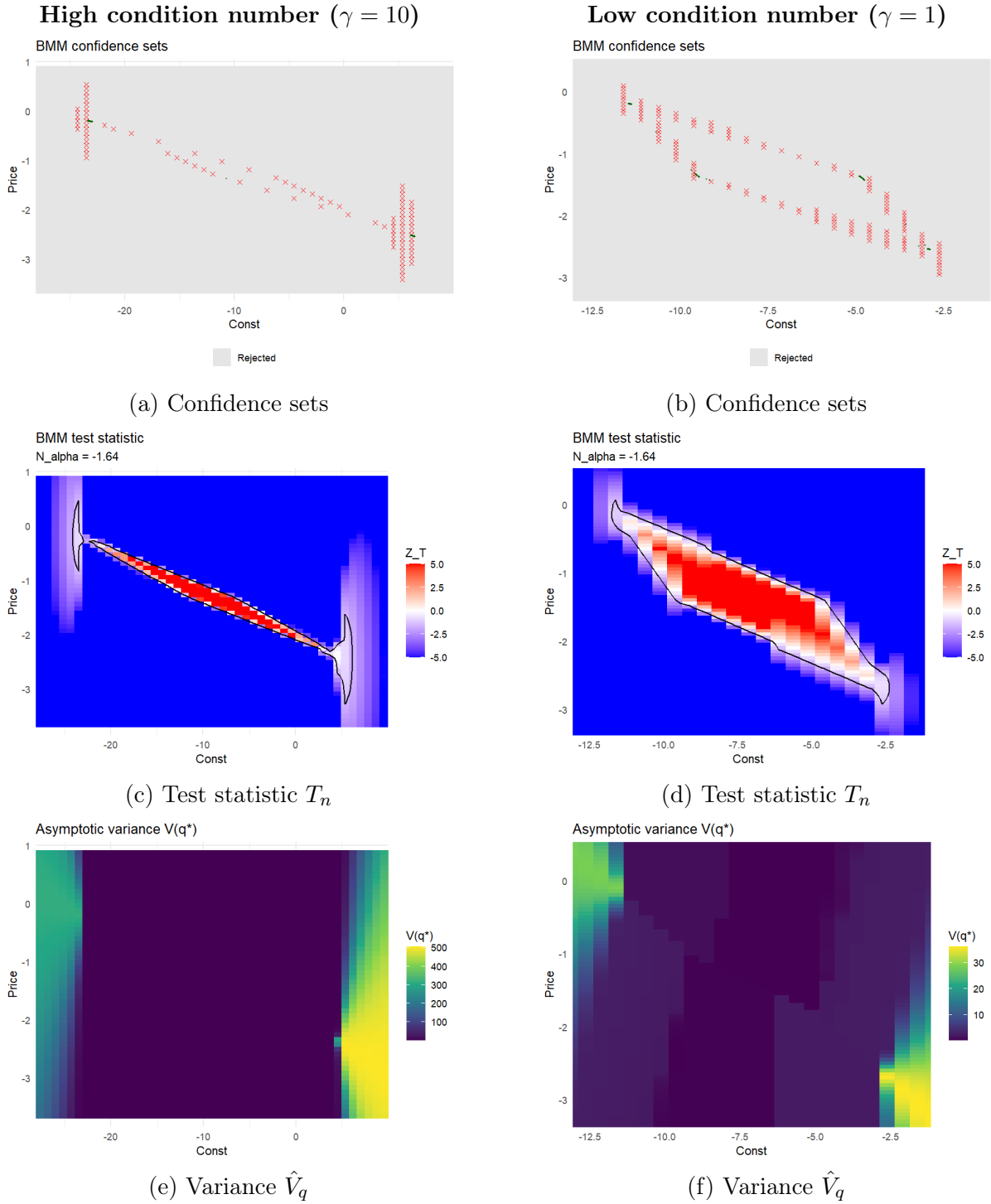


(d) Larger intervals; 100 markets; 5 products



Notes Boundary of the estimated identified set $\hat{\Theta}_{G,n}$ in the (Const, Price) plane for a single simulated dataset, traced by evaluating the empirical support function over 1,000 directions sampled from the unit sphere (green crosses). The solid black curve is the CCK identified-set frontier and the dashed blue lines mark the BMM projected identified intervals $[\hat{\theta}_j^L, \hat{\theta}_j^U]$. The red cross is the true parameter $\theta_0 = (-7, -1.5)$. All DGPs use a medium tail in the price distribution.

Figure 9: Effect of the first-stage condition number on BMM inference (medium tail, larger intervals; 100 markets; 5 products)



Notes Effect of the price constant shifter γ on the BMM inference, in the (Const, Price) plane for the larger-interval, 100-market design. X stacks the regressors (Const, Price). Z is the instrument matrix in saturated form: a constant together with the indicator dummies Z_{1jt}, Z_{2jt} (jittered by the CCMS smoothing), so Z has more columns than X . The BMM construction augments the regressors to $X_{\text{new}} = [X, Z_H]$, where Z_H is the orthonormalized part of the surplus instruments orthogonal to the orthonormalized fitted regressors Z_F (Z_F spans $\hat{X} = P_Z X$); this makes $Z'X_{\text{new}}$ square, and the frontier point $\beta_q^* = (Z'X_{\text{new}})^{-1}Z'\delta_q^*$ inverts exactly this matrix. Accordingly $\text{cond}(Z'X)$ denotes the condition number of $Z'X_{\text{new}}$ (the matrix BMM inverts), while $\text{cond}(X'P_Z X)$ is that of the 2×2 2SLS first-stage cross-product, with $P_Z = Z(Z'Z)^{-1}Z'$. Left column: $\gamma = 10$, a high condition number ($\text{cond}(X'P_Z X) = 8393.8$, $\text{cond}(Z'X) = 222.8$); right column: $\gamma = 1$, much lower (76.4 and 28.5). Rows show, from top: the BMM confidence sets of the identified set frontier, the studentized statistic $T_n(\theta_0)$, and the variance $\hat{V}_q$ across the grid of $\Theta_G(P)$. Under $\gamma = 10$ the intercept and price are nearly collinear after projecting onto Z , so $\hat{V}_q$ and T_n inflate along the Const direction and the confidence set fans out at the corners; under $\gamma = 1$ these artifacts vanish.

B. SUPPLEMENT ON DIGITAL APPS DATA ILLUSTRATION

Figure 10: Steam apps – Price distribution

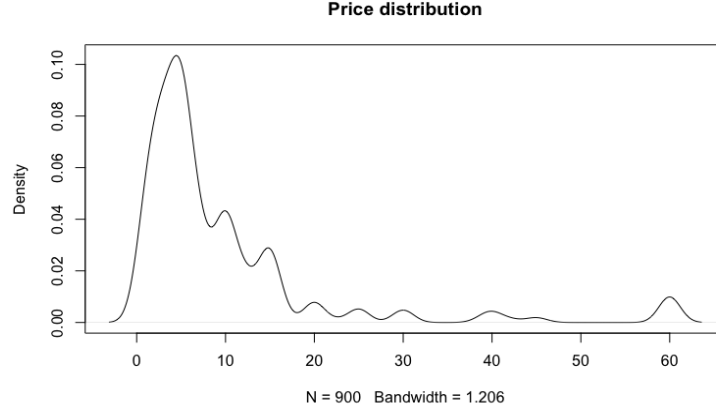


Figure 11: Steam data – Distribution of interval-valued base of users for all apps in Oct 2023

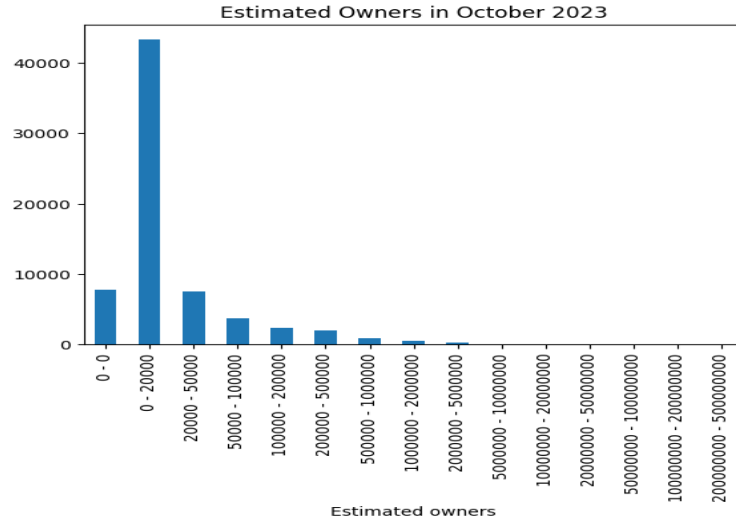


Figure 12: Steam apps – Distribution of Lower and Upper Bounds for δ_{jt}

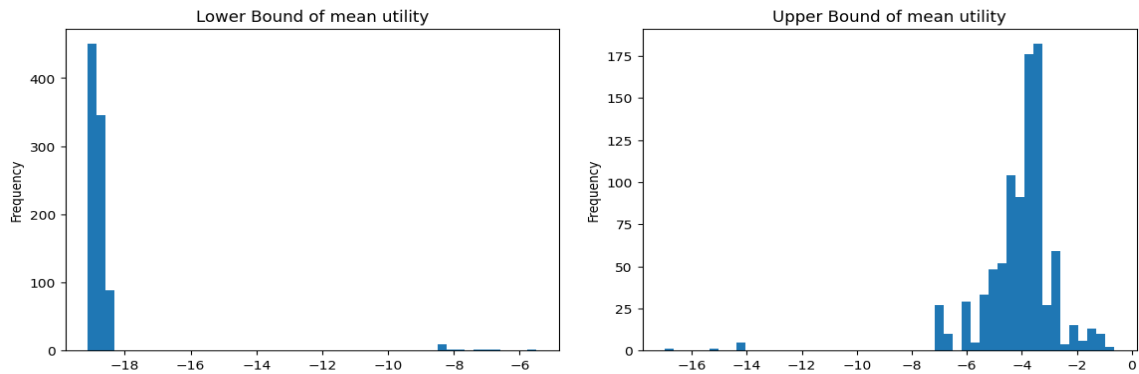


Table 10: Steam apps – Summary of demand data^(a)

Collect date	11-2023	12-2023	01-2024	02-2024	03-2024	04-2024	05-2024	06-2024	07-2024
Nb Apps	3937	4330	4594	4713	5317	5774	6216	6621	7070
Nb in outside option	3847	4240	4504	4623	5227	5684	6126	6531	6980
Nb in inside options	90	90	90	90	90	90	90	90	90
Inside options: casual and PTP apps^(b) from action-adventure genre, with at least 20K users reached in 07-2024)									
Avg Price	9.018	10.460	10.344	9.672	6.938	9.813	10.231	9.772	9.913
Free to play ratio	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Avg Middle for Base (1e5)	0.577	0.579	0.587	0.644	0.676	0.676	0.747	0.796	0.817
Avg Lower Bound for Base (1e5)	0.339	0.341	0.348	0.391	0.413	0.413	0.460	0.490	0.507
Avg Upper Bound for Base (1e5)	0.816	0.817	0.827	0.898	0.938	0.938	1.033	1.101	1.128
Avg Lower Bound for Sales (1e3)	0.333	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Avg Upper Bound for Sales (1e3)	45.778	47.778	48.556	55.000	54.667	52.444	62.000	64.111	63.778
Ratio of zero Lower Bound for Sales	0.989	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Outside option (apps from other major genres that do not satisfy criteria for inside option apps)									
Avg Price	8.383	7.943	7.981	6.244	7.949	8.078	8.371	8.183	8.183
FTP ratio	0.014	0.014	0.016	0.017	0.017	0.016	0.016	0.017	0.017
Avg Middle Base (1e5)	0.216	0.219	0.225	0.229	0.229	0.224	0.228	0.227	0.227
Avg Lower Bound for Base (1e5)	0.078	0.079	0.082	0.085	0.086	0.082	0.086	0.085	0.085
Avg Upper Bound for Base (1e5)	0.354	0.359	0.367	0.372	0.371	0.365	0.371	0.369	0.369
Avg Lower Bound for Sales (1e3)	1.175	0.330	0.180	0.344	0.631	0.167	0.424	0.214	0.202
Avg Upper Bound for Sales (1e3)	28.869	28.811	29.203	29.161	29.585	28.561	29.483	28.898	28.910
Ratio of zero Lower Bound for Sales	0.988	0.994	0.996	0.996	0.990	0.999	0.993	0.994	0.996

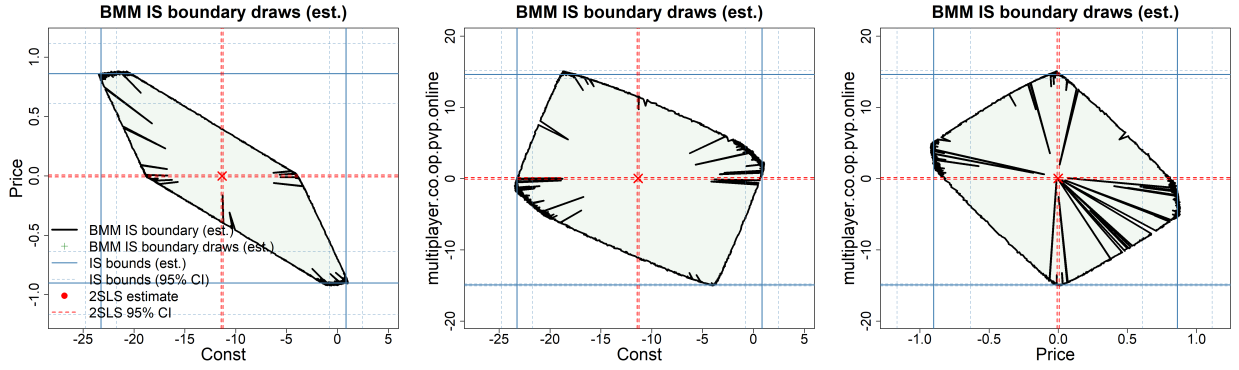
(a): Focus on recent apps (released since Q1-2023) with positive number of users and from major genres

(b): Pay to Play apps, i.e. apps that are accessible through some fees. Oppositely, free to play (FTP) apps have free access.

Table 11: Steam apps – Summary statistics on key variables

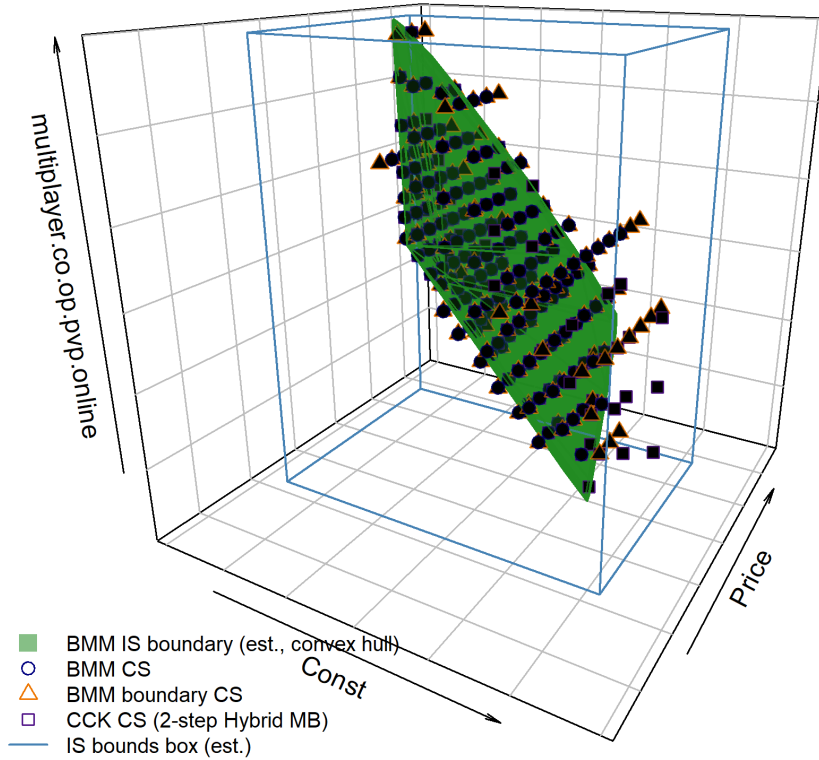
Variables	count	mean	std	min	25%	50%	75%	max
Lower Bound for Sales (1e3)	810	0.038	1.054	1e-3	1e-3	1e-3	1e-3	30
Upper Bound for Sales (1e3)	810	54.901	72.700	1e-3	30	30	50	80
Lower Bound for Sales of Outside good (1e5)	810	19.97778	11.6286	8.1	14	14.1	26	45.2
Upper Bound for Sales of Outside good (1e5)	810	1541.85556	298.30033	1110.6	1315.3	1546.4	1806.1	2017.9
Price (USD)	810	9.573	11.609	0.490	2.990	4.990	9.990	59.990
Instruments for price								
multiplayer/co-op/pvp/online	810	0.133	0.340	0	0	0	0	1
Log-Nb Releases	810	1.614	1.661	0	0	0.896	3.091	5.215
Single Release	810	0.344	0.475	0	0	0	1	1

Figure 13: Steam apps — 2D-Projections of the estimated identified set



Notes Boundary of the estimated identified set $\hat{\Theta}_{G,n}$ in the three coordinate projections, traced by evaluating the empirical support function over directions sampled from the unit sphere. The solid black curve connects the subsample of BMM-estimated identified-set boundary points sorted by polar angle around their centroid, forming an approximate frontier of the estimated identified set. Solid blue lines mark the estimated projected bounds $[\hat{\theta}_j^L, \hat{\theta}_j^U]$ and dashed blue lines their 95% confidence intervals. “Price” is the price coefficient θ_p and “multiplayer.co.op.pvp.online” is the multiplayer/online coefficient θ_{om} .

Figure 14: Steam apps – 95%-confidence sets, joint three-dimensional view



Notes The green surface is the convex hull of subsample of BMM-estimated identified-set boundary points; the blue box is the rectangle of estimated projected bounds. Markers are the non-rejected grid points for each test, as in Figure 3.

Table 12: Google PlayStore puzzle apps – summary statistics

	Mean	SD	Min	p25	Median	p75	Max
<i>Panel A: Sales bounds and app characteristics</i>							
Installs per day, lower S_L	693.29	2706.67	0.00	0.12	1.32	18.18	17825.31
Installs per day, upper S_U	2909.00	12657.12	0.01	0.61	6.62	90.91	89126.56
In-app purchases	0.51	0.50	0.00	0.00	1.00	1.00	1.00
Ad-supported	0.86	0.35	0.00	1.00	1.00	1.00	1.00
Non-zero rating	0.65	0.48	0.00	0.00	1.00	1.00	1.00
Log rating count per day	0.58	1.23	0.00	0.00	0.02	0.27	7.13
High rating count per day	0.33	0.47	0.00	0.00	0.00	1.00	1.00
Log size (MB)	3.32	0.86	0.18	2.83	3.37	3.87	6.15
Updated after release	0.78	0.41	0.00	1.00	1.00	1.00	1.00
More puzzle apps (developer)	0.62	0.48	0.00	0.00	1.00	1.00	1.00
<i>Panel B: Instrument inputs (developer aggregates and cross-platform gaps)</i>							
Installs, other puzzle apps	5.34	11.29	0.00	0.00	1.00	5.00	122.00
Installs, other non-puzzle apps	9.05	18.82	0.00	1.00	3.00	8.00	133.00
Log size, other puzzle apps	17.21	35.58	0.00	0.00	3.71	15.85	382.13
Log size, other non-puzzle apps	28.77	64.58	−1.50	1.86	8.29	28.17	482.50
Updated after release, other puzzle apps	4.29	9.30	0.00	0.00	1.00	4.00	104.00
Updated after release, other non-puzzle apps	6.92	15.40	0.00	0.00	2.00	6.00	127.00
Google–Apple log size gap (MB)	−1.19	0.81	−4.28	−1.63	−1.22	−0.75	2.93
Google–Apple age gap (days)	−95.56	512.94	−3996.00	−3.50	3.00	24.50	883.00

Notes Inside and outside goods are restricted to free apps that were released from 2018 to mid-2021 (the data collection period). There are $n = 799$ inside goods, i.e. free puzzle apps released in 2019, excluding the followings: apps flagged by Google editorial team, apps with no match on Apple store, apps with missing size, apps with uncoherent release and update dates, apps with ESRB $\neq$ Everyone. Sales and outside-option bounds are aggregate per-day installs: market size $M \in [10.46, 38.88]$ million, outside option $\in [9.90, 36.56]$ million, and the inside options (the 799 cohort apps) $\in [0.55, 2.32]$ million; these are constant across apps within the single market. Panel B variables enter the instrument set after a hypercube transformation and selection.

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